Index of every captured screen

655 screens, listed by the prototype they come from and the navigation that opens each one.

Prototype / template	Screens	Screens captured (by the control that opens them)
architecture	11	Opening view · Full stack & data · Read-model pipeline · AI gateway · Lag simulator · Migration phases · Full stack & data › Front end · Full stack & data › Middleware · Full stack & data › Server / API · Lag simulator › OffProjection worker failed · Lag simulator › OffBlue/green rebuild to tenant_overview_v2
atlas	16	Opening view · Applicant3 · Guardian3 · Alumni3 · Faculty3 · Teaching assistant1 · Advisor3 · Registrar3 · Finance3 · Student affairs2 · IT / security8 · Institution admin4 · Support / ops2 · Employer2 · Marketplace partner2 · Founder / company16
business	13	Opening view · Launch tracker · 90-day roadmap · Go-to-market · Launch checklist · Pilot checklist · Risk register · Hiring plan · Fundraising · Documents & decks · Launch tracker › P0 · Launch tracker › P1 · Launch tracker › P2
community	23	Opening view · Groups · Events · Q&A · Clubs & leaders · Mentorship · Marketplace · Messages · Privacy & safety · Groups › Open · Events › RSVP · Clubs & leaders › Leader tools · Clubs & leaders › Join · Clubs & leaders › Apply · Mentorship › Not for me · Marketplace › Bid $44 · Marketplace › Textbooks · Marketplace › Gear · Marketplace › Housing · Marketplace › Services · Marketplace › Tickets · Messages › Decline · Messages › Report or block
company-site	11	Opening view · Product · Students · Institutions · Pilot · Trust · Pricing · About · Talk to us · Legal · Pilot › Send
compliance	1	Opening view
console	8	Opening view · Approvals3 · Support queue27 · Integrations · Security operations · Audit log · Approvals3 › Approve… · Security operations › Break-glass access…
demo-video	1	Opening view
every-screen	19	Opening view · Public company, sales & marketing site80 · Faculty & Course Studio39 · Advisor & student success29 · Registrar & academic operations36 · Student accounts, aid & commerce29 · Campus life38 · Community & moderation35 · Family & guardian13 · Career, employer, alumni40 · Institutional administration49 · Integrations53 · Trust, privacy, security48 · Operations Command Center (/ops)102 · Loading · Empty · Error · Forbidden · Offline
external	5	Opening view · Guardian · Alumni · Employer · Marketplace partner
forms	3	Opening view · Capabilities · Interfaces
gap-screens	41	Opening view · Confirm identity · Connect SSO · Complete onboarding · Import or review course context · User asks · Intent identified · Prompt/tool permission validated · Response generated · Human escalation offered · Usage/spend logged · Feedback/correction captured · Publish course AI policy · Manage roster · Communicate with class · Refer to office · Review consented share · Conduct term review · Reconcile with SIS · Operational champion named · Reconciliation passed · Admin training · Student communication prepared · Launch readiness gate passed · Controlled launch · Adoption monitored · Midpoint review · Business purpose defined · Sandbox validated · Monitoring enabled · Write direction requested · Controlled write pilot · Rollback verified · Domain inventory · Validation · Dual run · User acceptance · Cutover · Enhanced monitoring · Rollback window · Legacy retirement
investor	6	Opening view · KPIs · Cash & runway · Pipeline · Risks · Data room
launch-kit	1	Opening view
master-catalog	9	Opening view · Roles · 51 · Screens · 673 · Workflows · 319 · Systems · 88 · Education graph · Documents · 82 · Gap register · Claude Code prompt
master-control	1	Opening view
mobile	6	Opening view · My Path · Search · Plan · Me · Me › OnFinal grades

Prototype / template	Screens	Screens captured (by the control that opens them)
one-place	8	Opening view · Student15 · Faculty & staff7 · Institution5 · Company (Semester)18 · Designed · To build · Embedded
ops	46	Opening view · Marketing & campaigns · Finance console · Sales & RevOps · Partnerships · Customer success · Support desk · Trust & safety · community · Legal & compliance · IT & security · People ops · Data & analytics · Marketing & campaigns › Registrar roundtable Events $61k 88 $693 Planning · Marketing & campaigns › Accessibility month Social · PR $22k 140 $157 Live · Marketing & campaigns › CIO webinar series Webinar $18k 206 $87 Live · Marketing & campaigns › Community college tour Field $96k 74 $1,297 Over budget · Finance console › New invoice · Finance console › Invoices & AR · Finance console › Contracts & renewals · Finance console › Usage billing · Finance console › AI cost · Finance console › Payroll summary · Sales & RevOps › Pilot → full Lakeshore College Proposal $210k Dec 2 L. Brandt · Sales & RevOps › 3 campuses Eastbrook University Procurement $880k Jan 20 R. Osei · Sales & RevOps › New logo Brightwater CC Discovery $95k Feb 28 M. Chen · Sales & RevOps › Expansion seats Valewood University Verbal $140k Oct 30 L. Brandt · Partnerships › LMS vendor B LMS Strategic Certified $1.1M Active · Partnerships › Regional reseller C Reseller Select — $0.7M Onboarding · Partnerships › Accessibility vendor D Tech Select In review $0.2M Pilot · Customer success › Ridgeway Community College B. Lowe 64 52% 5 Adoption plan · Customer success › Northmere University A. Ruiz 88 68% 1 Expansion call · Customer success › Coastline Tech K. Imani 41 29% 9 Exec save plan · Customer success › St. Aurel College B. Lowe 79 66% 0 — · Customer success › Valewood University K. Imani 85 70% 3 Training · Customer success › Pinecrest CC A. Ruiz 58 47% 6 Champion left · Customer success › Marrow Institute B. Lowe 81 63% 2 QBR Nov 4 · Customer success › Eastbrook University K. Imani 90 74% 1 Case study · Support desk › #58207 Coastline Tech SSO certificate rotation P2 2h left In progress · Support desk › #58199 Pinecrest CC Bulk course import CSV P3 1d Waiting on customer · Support desk › #58190 Northmere University Caption export format P4 3d Solved · Legal & compliance › New subprocessor notice AI vendor B Subprocessor GC Oct 18 30-day notice out · Legal & compliance › State privacy addendum Oakvale State Addendum GC Oct 25 Drafting · Legal & compliance › Records request Halden State Institution request Privacy Oct 9 Fulfilled · IT & security › CC7.2 Monitoring Ops Alert runbooks Platform — Passing · IT & security › CC8.1 Change mgmt Eng PR approvals sample P. Adeyemi — Passing · IT & security › A1.2 Backups Ops Restore test Platform Oct 30 Scheduled
platform-maturity	15	Opening view · Three tiers · 12 pillars · Scorecard · Competitive scorecard · APIs · Graph & authority · Roadmap · Moats & risks · Next 50 actions · Scorecard › 0 — None · Scorecard › 1 — Idea · Scorecard › 3 — Built, untested · Scorecard › 4 — In production with evidence · Scorecard › 5 — Proven at scale
role-lens	46	Opening view · Student64 · Applicant13 · Alumni13 · Faculty14 · Teaching assistant10 · Advisor11 · Registrar9 · Finance4 · Student affairs7 · IT / security4 · Support / ops3 · Employer4 · Marketplace partner2 · Department chair8 · Community moderator4 · Mentor6 · Implementation lead4 · Customer success1 · Revenue operations1 · Investor / board0 · Institution buyer2 · This role · This role › Course hub— none · This role › Notifications— none · This role › Registration— none · This role › Changes— none · This role › Decisions— none · This role › Runway— none · This role › Pathway— none · This role › Applying— none · This role › Launchpad— none · This role › Career— none · This role › Opportunities— none · This role › Costs & billing— none · This role › Notes & tasks— none · This role › Grades— none · This role › Import— none · This role › Sources— none · This role › Study— none · This role › Meet— none · This role › Ask Semester— none · This role › Analyse— none · This role › Solve— none · This role › Exam prep— none · This role › Work— none
role-pages	46	Opening view · Prospective student · Applicant · Transfer student · Online student · Adult learner · Graduate student · International student · Student athlete · Student worker · Student parent/caregiver · Peer mentor · Student organization leader · Family/guardian · Faculty/instructor · Teaching assistant · Department chair · Program director · Academic advisor · Student success staff · Registrar staff · Financial aid staff · Student accounts staff · Admissions staff · Housing staff · Dining/campus-card staff · Library staff · Accessibility/disability services · Counseling/wellbeing staff · Career staff · Campus safety staff · Community moderator · Institutional IT · Institutional security/privacy · Institutional executive · Institutional auditor · Alumni · Employer · Mentor · Developer/partner · Semester support operator · Semester implementation lead · Semester customer-success manager · Semester revenue operations · Semester finance operator · Semester security operator
staff	7	Opening view · TA grading queue16 · Advisees42 · Registration desk3 · Student accounts · Support requests2 · TA grading queue16 › Save draft and next

Prototype / template	Screens	Screens captured (by the control that opens them)
student	78	Opening view · App directory · Notifications · Calendar · Registration · Career · Costs & billing · Courses · Study · Ask Semester · Write · Sheets · Decks · Grades · Support · Family · Maps · Mail · Community · Me · Degree · Advising · Settings · Connect · Privacy & data · Search or run a command · App directory › Course hub · App directory › Changes · App directory › Decisions · App directory › Runway · App directory › Pathway · App directory › Applying · App directory › Launchpad · App directory › Opportunities · App directory › Notes & tasks · App directory › Import · App directory › Sources · App directory › Meet · App directory › Analyse · App directory › Solve · App directory › Exam prep · App directory › Work · App directory › Time · App directory › Group work · App directory › Draw · App directory › Equations · App directory › Create · App directory › Essay · App directory › Proofread · App directory › Catching up · App directory › People · App directory › University · App directory › Activities · App directory › Classmates · App directory › Dining · App directory › Housing · App directory › Links · App directory › Calls · App directory › Athletics · App directory › NIL · App directory › Reports · App directory › Account · App directory › Export · Registration › Review ENGL 245-03 · Registration › Review BIO 410-01 · Costs & billing › Pay $1,240.00… · Study › Show answer · Ask Semester › Add 3 blocks to calendar… · Decks › 2 Methods · Decks › 3 Results · Decks › 4 Discussion · Family › OnShare Final grades · Family › OnShare Billing milestones · Family › OffShare Class schedule · Mail › Registrar · Mail › Rowing club · Mail › Career Center · Advising › Preview and share…
wireframes	25	Opening view · Applicant13 · Guardian11 · Alumni13 · Faculty14 · Teaching assistant10 · Advisor11 · Registrar9 · Finance4 · Student affairs7 · IT / security4 · Support / ops3 · Employer4 · Marketplace partner2 · Department chair8 · Community moderator4 · Mentor6 · Implementation lead4 · Customer success1 · Revenue operations1 · Investor / board0 · Institution buyer2 · OnHide screens this role can't open · Open wireframe: Help · Open wireframe: University
workflow-lab	4	Opening view · Feature matrix · Benchmark lab · Benchmark lab › OnTested reload
workflow-runner	32	Opening view · Run a workflow · All workflows · By role · Screen coverage · Faculty course lifecycle24 · Advising & student success17 · Registrar & academic operations21 · Institution implementation22 · Controlled integration20 · Governed AI request17 · Support request12 · Incident response17 · Domain migration17 · Student finance & payment16 · Campus services15 · Community & moderation20 · Career, employer & alumni18 · Family & guardian consent13 · Developer platform & marketplace16 · Analytics, outcomes & reliability14 · Run a workflow › Auto-run · Run a workflow › Save · Run a workflow › Next → · All workflows › Start · By role › ◖ · Screen coverage › A · Public company, sales & marketing site · Advising & student success17 › Mark reviewed · Institution implementation22 › Attach evidence · Controlled integration20 › Send hand-off · Governed AI request17 › Run check · Domain migration17 › Run
Market Research	1	Opening view
Operating Manual	1	Opening view
Operating Pack	1	Opening view
Pilot Invitation Email	1	Opening view
Pilot One-Pager	1	Opening view
Vanderbilt Campus Map	1	Opening view
Data Hosting Map	1	Opening view
Incident Notice Email	1	Opening view
Semester Mark 3D	0	

Prototype / template	Screens	Screens captured (by the control that opens them)
Sponsor Weekly Update	1	Opening view
Staff Training Flier	1	Opening view
Welcome Email	1	Opening view
Semester Staff Workspaces	4	Opening view · Advisees · Registration desk · Student accounts
Command center screen	9	Opening view · Tenants · Revenue · Releases · Reliability · Support · Trust · Finance · Localisation
Company deck	14	Slide 1 · Slide 2 · Slide 3 · Slide 4 · Slide 5 · Slide 6 · Slide 7 · Slide 8 · Slide 9 · Slide 10 · Slide 11 · Slide 12 · Slide 13 · Slide 14
Institution console screen	1	Opening view
Education OS domain map	20	Opening view · Connect · Co-pilot · Domains · Core · OS · Productivity · AI · Advising · Degree audit · Campus services · Career · Family · Support · LMS · Gradebook · SIS · Registration · Student accounts · Company operations
Finish line	21	Opening view · Readiness · Release evidence · Cadence · 02 · 03 · 04 · 05 · 06 · 07 · 08 · 09 · 10 · 11 · 12 · 13 · 14 · 15 · 16 · 17 · 18
Institution buyer deck	5	Slide 1 · Slide 2 · Slide 3 · Slide 4 · Slide 5
Institution OS screen	7	Opening view · Communications · Course Studio · Registrar OS · Research · Admissions · Last year
Internal plan deck	4	Slide 1 · Slide 2 · Slide 3 · Slide 4
Investor deck	6	Slide 1 · Slide 2 · Slide 3 · Slide 4 · Slide 5 · Slide 6
Launch poster	1	Opening view
OnboardingAdvisor.dc.html	1	Opening view
OnboardingAlumni.dc.html	1	Opening view
OnboardingApplicant.dc.html	1	Opening view
OnboardingEmployer.dc.html	1	Opening view

Prototype / template	Screens	Screens captured (by the control that opens them)
OnboardingFaculty.dc.html	1	Opening view
OnboardingFinance.dc.html	1	Opening view
OnboardingGuardian.dc.html	1	Opening view
OnboardingITSecurity.dc.html	1	Opening view
OnboardingRegistrar.dc.html	1	Opening view
OnboardingStudent.dc.html	1	Opening view
OnboardingStudentAffairs.dc.html	1	Opening view
OnboardingSupport.dc.html	1	Opening view
OnboardingTA.dc.html	1	Opening view
Master operating map	21	Opening view · 1 · Education experience · 2 · Learning and assessment · 3 · Academic records and registrar · 4 · Student success and support · 5 · Productivity and knowledge · 6 · Governed AI · 7 · Campus life and commerce · 8 · Career and lifelong learning · 9 · Identity, trust and governance · 10 · Institutional intelligence · 11 · Platform and partners · 12 · Company operating system · 02 · Productivity and knowledge · 03 · Academic records and registrar · 04 · Learning and assessment · 05 · Learning and assessment · 06 · Governed AI · 07 · Productivity and knowledge · 08 · Education experience · 09 · Student success and support
Product overview deck	10	Slide 1 · Slide 2 · Slide 3 · Slide 4 · Slide 5 · Slide 6 · Slide 7 · Slide 8 · Slide 9 · Slide 10
Pilot results report	1	Opening view
Platform screen shell	7	Opening view · Study cards · Connections · Safety · Accessibility · Marketplace · Developers
Privacy center	1	Opening view
Product teaser (30 s)	1	Opening view

Prototype / template	Screens	Screens captured (by the control that opens them)
Student quick start	1	Opening view
Research brief	1	Opening view
Role onboarding	1	Opening view
Student screen	1	Opening view
Vision deck	12	Slide 1 · Slide 2 · Slide 3 · Slide 4 · Slide 5 · Slide 6 · Slide 7 · Slide 8 · Slide 9 · Slide 10 · Slide 11 · Slide 12

APPENDIX

Source research

The research reports and planning documents the founder gathered while building Semester, reproduced in full.

Map Out entire Semester OS

Semester OS is a **unified, AI-native operating system for education**: one platform that connects and progressively replaces fragmented student tools, learning systems, registrar/SIS functions, campus services, productivity suites, AI assistants, support systems, and institutional operations.

Its operating doctrine is:

> **Build core capabilities natively. Integrate responsibly during transition. Replace by domain only with evidence. Operate every domain as one coherent system.**

The full architecture is below.

Semester OS map

```
SEMESTER EDUCATION OPERATING SYSTEM
│
├── Public Company and Trust Layer
├── Identity, Tenant, and Policy Foundation
├── Student Daily Operating System
├── Productivity and Collaboration Workspace
├── Academic Path and Planning System
├── Course Studio and Learning System
├── Governed AI and Intelligence Platform
├── Advising, Student Success, and Support System
├── Registrar, SIS, and Academic Records System
├── Student Finance and Commerce System
├── Campus Life and Service Super-App
├── Community and Social Connection System
├── Career, Alumni, and Lifelong Learning System
├── Family and Guardian Consent System
├── Institutional Administration and Governance System
├── Integration, Migration, and Interoperability Fabric
├── Security, Privacy, Accessibility, and Compliance System
├── Operations Command Center
├── GTM, Revenue, Billing, and Customer Success System
├── Developer Platform and Marketplace
└── Analytics, Outcomes, Reliability, and Continuous Improvement
```

Shared foundation

Every module shares the same control chain:

```
Identity
  ↓
Institution / tenant
  ↓
Membership
  ↓
Role
  ↓
Capability
  ↓
Scope
  ↓
Policy
  ↓
Data classification
  ↓
Consent
  ↓
Source authority
  ↓
Workflow
  ↓
Audit
  ↓
Evidence
  ↓
Outcome
```

This is what prevents Semester from becoming a collection of disconnected apps.

Core user journey

```
Student signs in
  ↓
Today: sees meaningful current context
  ↓
Action Center: sees what matters now
  ↓
Path: understands degree/program progress
  ↓
Plan: compares and plans terms/courses
  ↓
Courses: learns, studies, submits work, receives feedback
  ↓
Copilot: gets source-aware, policy-bounded help
  ↓
Support: reaches the correct office or person
  ↓
Campus: uses student-life services
```

```
  ↓
Career: builds skills, evidence, credentials, and opportunities
  ↓
Lifelong relationship: alumni, mentoring, continuing education
```

Student OS

```
Student Home
├── Today
├── Action Center
├── Calendar
├── Schedule
├── Tasks
├── Goals
├── Path
├── Degree progress
├── Program requirements
├── Term planning
├── Registration readiness
├── Course comparison
├── Workload planning
├── Cost planning
├── Courses
├── Notes
├── Files
├── Workspace
├── Search
├── AI Copilot
├── Community
├── Campus
├── Support
├── Career
├── Student account
├── Profile
├── Privacy settings
├── AI settings
├── Notifications
└── Data export/delete controls
```

Student daily loop

```
What changed?
  ↓
What matters?
  ↓
What can I do?
  ↓
What is official?
  ↓
Who can help?
  ↓
What progress did I make?
```

Productivity workspace

```
Workspace
├── Tasks
├── Calendar
├── Notes
├── Documents
├── Files
├── Projects
├── Group workspaces
├── Research workspace
├── Writing workspace
├── Focus mode
├── Study sessions
├── Templates
├── Search
├── Notifications
└── Automations
```

Every task, note, file, or event can connect to:

```
Course
Assignment
Learning objective
Program requirement
Academic term
Advisor
Support office
Career competency
Institutional deadline
Student goal
```

Academic path system

```
Academic Path
├── Institution
├── Program
├── Major/minor
├── Degree requirements
├── Course history
├── Current term
├── Future terms
├── Degree audit
├── Graduation scenarios
├── Transfer credits
├── Articulation rules
├── Prerequisites
├── Holds
├── Registration eligibility
├── Workload plan
```

```
├── Cost plan
└── Official record handoff
```

Source labels

```
Official institutional record
Semester planning scenario
Student-entered plan
External/transferred record
AI explanation
Advisor/registrar confirmation required
```

Course Studio and Learning OS

```
Course Studio
├── Course home
├── Syllabus
├── Learning objectives
├── Schedule
├── Announcements
├── Assignments
├── Assessments
├── Rubrics
├── Resources
├── Readings
├── Annotations
├── Study packs
├── Flashcards
├── Practice
├── Discussions
├── Groups
├── Office hours
├── Feedback
├── Gradebook
├── Regrades
├── Course AI rules
├── Academic integrity
├── Accessibility support
├── Learning evidence
└── Course analytics
```

Learning flow

```
Faculty publishes course structure
  ↓
Student receives course context
  ↓
Student studies and completes work
  ↓
Semester connects tasks, calendar, notes, resources, and AI support
  ↓
```

```
Faculty reviews and provides feedback
  ↓
Grade/feedback is released
  ↓
Learning evidence and path progress update
```

Governed AI platform

```
Semester AI Gateway
├── Student Copilot
├── Course Tutor
├── Research Assistant
├── Writing Assistant
├── Study Planner
├── Registration Guide
├── Academic Path Explainer
├── Advisor Assistant
├── Faculty Assistant
├── Staff Workflow Assistant
├── Institutional Knowledge Assistant
├── Model Registry
├── Provider Registry
├── Policy Engine
├── Course AI Rules
├── Data Classification
├── Consent Controls
├── Source Retrieval
├── Citation/Provenance
├── Tool Permissions
├── Usage Quotas
├── Spend Meter
├── Evaluation Harness
├── Safety/Red-Team Tests
├── Human Escalation
└── AI Audit Log
```

AI workflow

```
User request
  ↓
Role / tenant / course context
  ↓
Policy + consent + classification check
  ↓
Approved provider/model routing
  ↓
Approved source retrieval
  ↓
AI response with source/provenance
  ↓
Safe suggested action or official handoff
```

```
  ↓
Usage, feedback, audit, evaluation
```

AI can explain an official rule. It cannot silently become the official rule.

Advising, student success, and support

```
Student Success OS
├── Caseload
├── Student path
├── Action completion
├── Check-ins
├── Appointments
├── Success plans
├── Advising notes
├── Referrals
├── Outreach
├── Support requests
├── Student-consented shares
├── Intervention tracking
├── Cohort view
├── Program outcomes
├── Escalation
└── Aggregate insight
```

Support flow

```
Student need
  ↓
Self-service answer or support request
  ↓
Safe context and correct routing
  ↓
Office/team ownership
  ↓
Response, resolution, or referral
  ↓
Student status update
  ↓
Outcome/quality signal
```

Registrar, SIS, and academic operations

```
Registrar OS
├── Academic calendar
├── Terms
├── Catalog
├── Courses
├── Sections
├── Faculty assignments
├── Meeting times
```

```
├── Rooms
├── Capacity
├── Waitlists
├── Prerequisites
├── Registration windows
├── Time tickets
├── Holds
├── Overrides
├── Enrollment
├── Add/drop
├── Withdrawal
├── Academic record ledger
├── Grade ledger
├── Grade changes
├── Regrades
├── Degree audit
├── Transfer evaluation
├── Articulation rules
├── Graduation checks
├── Credential conferral
├── Transcript/export
├── Reconciliation
└── Migration center
```

Registration workflow

```
Student selects section
  ↓
Term, hold, prerequisite, capacity, credit, and policy checks
  ↓
Enroll / waitlist / deny / route for review
  ↓
Immutable audit entry
  ↓
Student receives authoritative result
```

Student finance and campus commerce

```
Student Finance OS
├── Student account
├── Charges
├── Credits
├── Balance
├── Holds
├── Payment plan
├── Installments
├── Payment handoff
├── Refund status
├── Scholarships
├── Financial-aid handoff
├── Cost planner
├── Budget planner
├── Emergency support
```

```
├── Reconciliation
└── Audit ledger
```

Campus super-app

```
Campus OS
├── Dining
│   ├── Locations
│   ├── Hours
│   ├── Menus
│   ├── Meal plans
│   ├── Balances
│   ├── Orders
│   └── Donation pool
├── Housing
├── Events
├── Organizations
├── Study spaces
├── Transit
├── Parking
├── Recreation
├── Library
├── Bookstore
├── Campus jobs
├── Volunteer opportunities
├── Local services
├── Safety resources
├── Wellness handoffs
└── Accessibility services
```

Community and social connection

```
Community OS
├── Community directory
├── Community feed
├── Posts
├── Comments
├── Reactions
├── Groups
├── Study sessions
├── Peer mentoring
├── Organization spaces
├── Alias/pseudonym controls
├── Reporting
├── Moderation
├── Appeals
├── Safety escalation
├── Community retention
└── Audit
```

Career, alumni, employer, and lifelong learning

```
Career OS
├── Skills graph
├── Evidence portfolio
├── Verified skills
├── Credential wallet
├── Resume
├── Cover letters
├── Interview prep
├── Opportunities
├── Internships
├── Employer profiles
├── Applications
├── Mentors
├── Alumni network
├── Career appointments
├── Career events
├── Continuing education
├── Microcredentials
├── Credential renewal
└── Career outcomes
```

Family and guardian consent system

```
Family Support OS
├── Consent invitation
├── Relationship confirmation
├── Scoped grants
├── Chosen information categories
├── Time-limited access
├── Shared items
├── Billing/support handoff
├── Communication preferences
├── Access history
├── Revocation
└── Audit
```

Student control is mandatory:

```
Student chooses scope
→ family accepts
→ only permitted information is visible
→ access is audited
→ access expires or can be revoked
```

Institutional control plane

```
Institution OS
├── Tenant overview
├── Campus hierarchy
├── SSO
├── SCIM
├── Membership
├── Roles
├── Capabilities
├── Access reviews
├── Plans and entitlements
├── Feature policies
├── Cohorts
├── Kill switches
├── Module modes
├── Rollouts
├── Configuration Studio
├── Workflow Builder
├── Policy hierarchy
├── Governance stewards
├── Governance decisions
├── Integrations
├── Data classification
├── Consent policy
├── Retention
├── Legal holds
├── Data requests
├── AI provider policy
├── Approved sources
├── Compliance
├── Trust artifacts
├── Accessibility program
├── Incident notices
├── Risks
├── Audit explorer
├── Offboarding
└── Portability/export
```

Integration and migration fabric

```
Integration Hub
├── Provider directory
├── Connections
├── Security/privacy review
├── Scope approval
├── Mapping Studio
├── Mapping versions
├── Transform rules
├── Sync schedule
├── Sync runs
├── Sync errors
├── Dead-letter queue
```

```
├── Replay requests
├── Reconciliation
├── Schema drift
├── Duplicate resolution
├── Source ownership
├── Freshness monitoring
├── Webhooks
├── LTI
├── OneRoster
├── Edu-API
├── SIS/ERP adapters
├── LMS adapters
├── Google/Microsoft/Apple adapters
├── AI-provider adapters
└── Integration retirement
```

Migration model

```
Inventory
→ map
→ stage
→ validate
→ reconcile
→ dual run
→ user acceptance
→ institutional approval
→ cutover
→ monitor
→ rollback window
→ archive/export
→ legacy retirement
```

Trust, privacy, security, and compliance

```
Trust System
├── Trust Center
├── Security overview
├── Privacy overview
├── Accessibility statement
├── Responsible AI
├── Data practices
├── Subprocessors
├── Reliability/status
├── HECVAT readiness
├── DPA process
├── Security questionnaires
├── Trust room
└── Vulnerability reporting

Internal Trust and Security
├── Asset inventory
├── Access review
├── Secret inventory
```

```
├── Vulnerabilities
├── Threat models
├── Security alerts
├── Incidents
├── Controls
├── Evidence
├── Vendor risk
├── Pen-test evidence
├── Break-glass review
├── Backup/restore
├── Disaster recovery
├── Data inventory
├── Classification
├── Retention
├── Legal holds
├── Privacy requests
├── Erasure/export
└── Audit
```

Operations Command Center

```
/ops
├── Executive Overview
├── Operations Inbox
├── My Work
├── Approvals
├── Tenants
├── Tenant 360
├── Customers
├── Customer 360
├── GTM Accounts
├── Pilots
├── Implementation
├── Success Plans
├── QBRs
├── Renewals
├── Customer Health
├── Revenue
├── Billing
├── Support
├── Product Operations
├── Releases
├── Feature flags
├── Beta
├── Integrations
├── Trust
├── Compliance
├── Privacy
├── Security
├── Access Reviews
├── Break Glass
├── Incidents
├── SLOs
├── Error Budgets
├── Audit Explorer
```

```
├── Risks
├── Vendors
├── Finance
├── Forecasts
├── Reports
├── Board/Investor
├── Capacity
└── Preferences
```

Operations action model

```
Request
→ attach evidence
→ required approval
→ impact preview
→ execution
→ audit event
→ projection/read-model refresh
→ operator review
```

Company operating system

```
Semester Company OS
├── Corporate governance
├── Legal operations
├── Finance
├── Budget/runway
├── Pricing
├── Sales
├── Marketing
├── GTM
├── CRM
├── Pilots
├── Customer success
├── Implementation
├── Support
├── Billing
├── Collections
├── Partnerships
├── Marketplace
├── Vendor management
├── Product operations
├── Engineering
├── Reliability
├── Security
├── Privacy
├── Compliance
├── AI governance
├── People/hiring
├── Board/investor reporting
├── Risk management
```

```
├── Release operations
└── Operating cadence
```

Developer platform and marketplace

```
Developer Platform
├── API documentation
├── OAuth/OIDC
├── API keys
├── Service accounts
├── Webhooks
├── Event subscriptions
├── SDKs
├── Sandbox tenants
├── Test data
├── Integration certification
├── Marketplace
├── Partner directory
├── App review
├── Data scopes
├── Versioning
├── Metering
├── Billing
├── Support
└── Incident notices
```

Required shared workflows

```
Authentication
Tenant provisioning
Role/capability grant
Consent grant/revocation
Course enrollment
Task creation/completion
Academic plan update
Registration request
Grade entry/release
Support request
Advisor referral
AI request
Integration sync
Data export/erasure
Feature rollout
Tenant rollout
Pilot conversion
Billing/payment
Incident response
Domain migration
Offboarding
```

Completion definition

No screen, workflow, system, or domain is complete until it has:

```
1. Product requirement
2. Role definition
3. Route/screen
4. Data model
5. Source authority
6. Tenant scope
7. Permission/capability
8. Data classification
9. Consent model
10. Accessibility evidence
11. Security controls
12. Privacy/retention rules
13. Audit event
14. Integration plan
15. Error/loading/empty/forbidden state
16. Support path
17. Monitoring/SLO
18. Test coverage
19. Rollback plan
20. Documentation
21. Release evidence
22. Owner and review date
```

Semester OS in one flow

```
Prospect
→ applicant
→ student
→ learner
→ planner
→ course participant
→ community member
→ support recipient
→ credential holder
→ job candidate
→ employee
→ alumnus
→ lifelong learner
→ mentor / partner / employer

Institution
→ configures
→ provisions
→ governs
→ supports
→ measures
→ improves
→ trusts
→ expands
```

```
→ migrates
→ operates
```

Master command

```
Create the definitive Semester OS master map.

Audit the Semester repository, database, migrations, routes, screens,
components, operations console, company site, integrations, policies, tests,
and documentation.

Produce a complete catalog of:

- Every user role
- Every public page
- Every student screen
- Every faculty screen
- Every advisor/student-success screen
- Every registrar/academic operations screen
- Every finance/student-account screen
- Every campus service screen
- Every community/moderation screen
- Every family/guardian screen
- Every career/employer/alumni screen
- Every institutional admin/control-plane screen
- Every integration/migration screen
- Every trust/security/privacy/compliance screen
- Every operations-console screen
- Every GTM/customer-success/billing/company operation screen
- Every developer/marketplace screen
- Every workflow
- Every system
- Every data domain
- Every graph relationship
- Every integration
- Every required document
- Every release gate
- Every test gate
- Every risk
- Every owner
- Every P0/P1/P2/P3 gap

Create:

docs/master/SEMESTER_OS_MASTER_MAP.md
docs/master/SEMESTER_ROLE_CATALOG.md
docs/master/SEMESTER_SCREEN_CATALOG.md
docs/master/SEMESTER_WORKFLOW_CATALOG.md
docs/master/SEMESTER_SYSTEM_CATALOG.md
docs/master/SEMESTER_DATA_DOMAIN_CATALOG.md
docs/master/SEMESTER_EDUCATION_GRAPH.md
docs/master/SEMESTER_CONSOLE_CATALOG.md
docs/master/SEMESTER_DOCUMENT_CATALOG.md
docs/master/SEMESTER_INTEGRATION_CATALOG.md
docs/master/SEMESTER_RELEASE_GATE_CATALOG.md
```

```
docs/master/SEMESTER_TRACEABILITY_MATRIX.md
docs/master/SEMESTER_GAP_REGISTER.md

For every item record:
- name
- domain
- role
- route
- job to be done
- source authority
- data model
- tenant scope
- permission/capability
- consent requirement
- AI policy
- accessibility requirement
- security/privacy/retention requirements
- audit event
- integration/migration dependency
- support owner
- SLO/reliability class
- tests
- release gate
- evidence
- current status
- risk
- priority
- next action

Do not mark anything complete unless code, permissions, RLS/tenant isolation,
accessibility, tests, monitoring, support, documentation, and release evidence
all exist.
```

Map semester out as a whol provide full map out of every workflow and role function screen system of semester

Semester should be mapped as one **Education Operating System** with shared identity, policy, data authority, audit, and workflow foundations—not as hundreds of disconnected screens. Below is the master map of roles, systems, screens, workflows, data domains, consoles, and company functions that make up Semester.

The central rule is:

> **Every Semester capability must connect to identity, tenant, permissions, policy, source authority, audit, support, and measurable outcomes.**

This map reflects Semester's full product thesis: student OS, productivity suite, LMS/Course Studio, registrar/SIS pathway, advising and student success, campus super-app, governed AI, marketplace/platform, institutional control plane, and company operating system. The existing architecture materials describe the same direction: build native capabilities, integrate during transition, replace only after evidence, and operate all domains as one system.[1]

Semester master map

```
SEMESTER EDUCATION OPERATING SYSTEM
│
├── 1. Public Company and Growth Layer
├── 2. Identity, Tenant, Policy, and Trust Foundation
├── 3. Student OS
├── 4. Productivity and Collaboration Workspace
├── 5. Academic Path and Degree Planning
├── 6. Course Studio and Learning OS
├── 7. Governed AI and Intelligence Platform
├── 8. Advising, Student Success, and Support
├── 9. Registrar, SIS, and Academic Operations
├── 10. Student Accounts, Billing, and Financial Workflows
├── 11. Campus Life and Service Super-App
├── 12. Community, Organizations, and Peer Connection
├── 13. Career, Alumni, Employer, and Lifelong Learning
├── 14. Family and Guardian Consent Platform
├── 15. Institutional Administration and Control Plane
├── 16. Integration, Migration, and Interoperability Fabric
├── 17. Security, Privacy, Accessibility, Compliance, and Trust
├── 18. Operations Command Center
├── 19. Revenue, GTM, Customer Success, and Company Operations
```

```
├── 20. Developer Platform, Marketplace, and Partner Ecosystem
└── 21. Analytics, Outcomes, Reliability, and Continuous Improvement
```

Shared foundation

Every system runs through the same shared control model:

```
Identity
  ↓
Tenant / institution scope
  ↓
Membership
  ↓
Role
  ↓
Capability
  ↓
Purpose
  ↓
Policy
  ↓
Data classification
  ↓
Consent / grant
  ↓
Source authority
  ↓
Workflow
  ↓
Audit event
  ↓
Evidence
  ↓
Outcome
```

Shared data objects

```
Person
Institution
Campus
School
Department
Program
Term
Course
Section
Enrollment
Role grant
Capability
Policy
Consent
Plan
Action
```

```
Task
Assignment
Resource
Evidence
Skill
Credential
Support request
Appointment
Community
Organization
Opportunity
Contract
Tenant
Customer
Integration
Audit event
Incident
Decision
Workflow
```

1. Public company and growth layer

Users

- Prospective student
- Prospective institutional buyer
- Student-success leader
- Advisor
- Registrar
- CIO/IT/security reviewer
- Procurement/legal/privacy reviewer
- Parent/family visitor
- Employer/partner
- Candidate/employee
- Press/media
- Investor
- Existing customer
- Developer/integration partner

Screens

```
/
├── Homepage
├── Product overview
├── Student workspace page
```

```
├── Institution platform page
├── Registration Readiness Pilot page
├── Course Studio page
├── Governed AI page
├── Student Success page
├── Registrar/Academic Operations page
├── Campus Services page
├── Career/Lifelong Learning page
├── Integrations page
├── Solutions
│   ├── Students
│   ├── Advisors
│   ├── Faculty
│   ├── Departments
│   ├── Registrars
│   ├── Student Success
│   ├── Institutions
│   └── Enterprise/Multi-campus
├── Pilot
├── Pricing
├── Feature comparison
├── Request demo
├── Request pricing
├── Trust Center
├── Security
├── Privacy
├── Accessibility
├── Responsible AI
├── Resources
├── Guides
├── Webinars
├── Research
├── Product updates
├── Case studies
├── Compare pages
├── Partners
├── Ambassadors
├── Careers
├── Company
├── Contact
├── Legal
├── Login
├── Sign up
├── Help handoff
├── Status handoff
└── 404 / maintenance
```

Workflows

```
Institutional visitor
→ pilot/use-case page
→ demo request
→ consent and attribution capture
→ CRM/GTM routing
→ discovery call
```

```
→ pilot design
→ security/procurement
→ paid pilot
→ implementation
→ annual conversion

Student visitor
→ student value page
→ product walkthrough
→ sign up
→ onboarding
→ first meaningful action
→ retained use
→ referral/Premium conversion
```

2. Identity, tenant, policy, and trust foundation

Roles

- Student
- Faculty
- TA
- Advisor
- Registrar
- Institutional admin
- Security admin
- Privacy officer
- IT administrator
- Support operator
- Finance operator
- Semester executive
- Auditor
- Partner developer
- Service account
- Family/guardian
- Employer
- Moderator

Screens

```
Identity and Access
├── Sign in
├── Sign up
├── Password reset
├── Magic link / MFA
├── SSO launch
├── Account linking
├── Profile
├── Account settings
├── Privacy settings
├── Notification settings
├── Security settings
├── Connected accounts
├── Institution membership
├── Role view
├── Capability view
├── Data export
├── Data request
├── Account deletion request
└── Session/device management
```

Core workflows

```
Account creation
→ identity verification
→ invite/SSO eligibility
→ tenant membership
→ role grant
→ capability resolution
→ onboarding
→ audit

Institution provisioning
→ identity provider configuration
→ SCIM or roster provisioning
→ membership reconciliation
→ role assignment
→ access review
→ deprovisioning
```

Core systems

```
Authentication
SSO
SCIM
Role grants
Capabilities
Tenant isolation
RLS
Approval workflow
```

```
Break-glass access
Consent
Data classification
Retention
Legal holds
Audit
Feature flags
Rollout policies
```

3. Student OS

Student screens

```
Student Home
├── Today
├── Action Center
├── My Path
├── My Plan
├── Calendar
├── Schedule
├── Tasks
├── Goals
├── Notes
├── Files
├── Workspace
├── Search
├── Courses
├── Study
├── Community
├── Support
├── Campus
├── Career
├── Student Account
├── Settings
└── AI Copilot
```

Student daily workflow

```
Sign in
→ see Today
→ understand urgent action
→ review context/source
→ complete action or request help
→ update plan/progress
→ return to dashboard with current status
```

Student action types

```
Academic
- Registration
- Course selection
- Assignment
- Study plan
- Degree requirement
- Grade/regrade

Support
- Advising
- Tutoring
- Financial support
- Accessibility
- Housing/dining
- Wellness resource handoff

Career
- Resume
- Internship
- Mentor
- Opportunity
- Portfolio

Campus
- Event
- Organization
- Dining
- Study space
- Campus job
```

4. Productivity and collaboration workspace

Screens

```
Workspace
├── My tasks
├── Shared tasks
├── Calendar
├── Notes
├── Documents
├── Files
├── Projects
├── Templates
├── Focus mode
├── Study sessions
├── Group workspaces
├── Research workspace
├── Writing workspace
├── Meeting notes
├── Notifications
```

```
├── Search
└── Workspace settings
```

Workflows

```
Create task
→ connect to course/goal/action
→ assign due date and priority
→ complete or defer
→ progress recorded
→ analytics/action center updates

Create note
→ link to course/resource/assignment
→ annotate
→ search later
→ optionally share with group/advisor
→ preserve owner/permission/audit boundaries

Create group workspace
→ invite permitted members
→ assign roles
→ create project/tasks/files
→ collaborate
→ close/archive/export
```

5. Academic path and degree planning

Screens

```
Academic Path
├── Program overview
├── Degree requirements
├── Requirement detail
├── Course history
├── Current term
├── Future-term plan
├── Graduation scenarios
├── Major/minor comparison
├── Degree audit
├── Transfer credit
├── Articulation rules
├── Workload planner
├── Cost planner
├── Academic standing
├── Holds
├── Registration readiness
└── Official-record handoff
```

Workflow

```
Student selects goal
→ Semester loads official/authorized program data
→ shows requirements and course history
→ calculates planning scenario
→ marks official versus estimated result
→ student adjusts plan
→ Semester identifies next action
→ official system handoff or native transaction
```

Required authority labels

```
Official institution record
Semester planning scenario
Student-entered plan
External/transferred record
AI explanation
Requires advisor/registrar confirmation
```

6. Course Studio and Learning OS

Student screens

```
Course Home
├── Syllabus
├── Objectives
├── Announcements
├── Schedule
├── Assignments
├── Resources
├── Readings
├── Notes
├── Study packs
├── Flashcards
├── Practice
├── Discussions
├── Groups
├── Office hours
├── Feedback
├── Grades
├── Regrades
├── Course AI rules
├── Course Copilot
└── Course settings
```

Faculty screens

```
Faculty Course Studio
├── Course setup
├── Syllabus editor
├── Objectives
├── Course guidance
├── Course AI policy
├── Approved sources
├── Resource library
├── Assignment builder
├── Assessment builder
├── Question bank
├── Rubric builder
├── Gradebook
├── Grade moderation
├── Grade release
├── Regrade queue
├── Feedback workspace
├── Study packs
├── Office hours
├── Group management
├── TA management
├── Course analytics
├── Accessibility review
├── Integrity workflow
├── LTI settings
└── Archive/export
```

Learning workflow

```
Faculty publishes:
syllabus + objectives + resources + AI policy
  ↓
Student enters course
  ↓
Student sees tasks, due dates, resources, guidance
  ↓
Student studies, creates notes, practices, asks policy-bounded AI
  ↓
Student submits work / completes assessment
  ↓
Faculty reviews, grades, gives feedback
  ↓
Grade is released or passed back under controlled policy
  ↓
Learning evidence and course progress update
```

7. Governed AI and intelligence platform

AI user experiences

```
Student AI Copilot
Course Copilot
Research Assistant
Writing Assistant
Study Planner
Path and Registration Guide
Advisor Assistant
Faculty Authoring Assistant
Staff Workflow Assistant
Institutional Knowledge Assistant
Support Assistant
Operator Assistant
```

AI screens

```
AI Copilot
├── New conversation
├── Conversation history
├── Context selector
├── Course context
├── Source panel
├── Citation panel
├── Suggested actions
├── Policy notice
├── Usage/budget view
├── AI settings
├── Memory controls
├── Feedback
├── Report problem
└── Human handoff
```

AI workflow

```
User request
→ identity and tenant identified
→ role/capability checked
→ course/tenant policy checked
→ data classification checked
→ consent checked
→ provider/model selected
→ approved sources retrieved
→ answer generated
→ citations/source authority shown
→ action suggestion/handoff offered
→ usage, audit, feedback, and evaluation signal recorded
```

AI governance systems

```
Model registry
Provider registry
AI policy
Course AI rules
Data classification
Prompt registry
Tool permission matrix
Source/provenance
Evaluation harness
Safety/red-team suite
AI incident response
Cost/spend meter
Quota/rate limit
Human override
Human escalation
Audit log
```

8. Advising, student success, and support

Advisor screens

```
Advisor Workspace
├── Caseload
├── Student profile
├── Path view
├── Academic plan
├── Action completion
├── Holds/status
├── Check-ins
├── Appointments
├── Advising notes
├── Success plans
├── Referrals
├── Student-consented shares
├── Outreach queue
├── Cohort view
├── Program outcomes
├── Intervention tracker
├── Escalation queue
└── Settings
```

Student-support screens

```
Support Center
├── Search help
├── Resource directory
├── Open support request
├── My support requests
├── Request detail
├── Messages
```

```
├── Referral status
├── Appointment scheduling
├── Emergency resource handoff
├── Accessibility support
├── Financial support
└── Feedback
```

Workflow

```
Student identifies need
→ finds self-service answer or opens request
→ safe context attached
→ correct office receives request
→ owner accepts/routes/responds
→ student sees progress
→ issue resolved or escalated
→ outcome recorded
→ aggregate service insight improves operations
```

9. Registrar, SIS, and academic operations

Screens

```
Registrar Console
├── Terms
├── Academic calendar
├── Catalog
├── Courses
├── Sections
├── Faculty assignments
├── Meeting times
├── Rooms
├── Capacity
├── Waitlists
├── Prerequisites
├── Co-requisites
├── Registration windows
├── Time tickets
├── Holds
├── Overrides
├── Enrollment queue
├── Add/drop/withdrawal
├── Registration audit
├── Academic records
├── Record corrections
├── Grade ledger
├── Grade changes
├── Regrades
├── Degree audit rules
├── Transfer evaluations
├── Articulation rules
├── Graduation review
```

```
├── Credential conferral
├── Transcript/export
├── Reconciliation
├── Migration center
└── Settings
```

Registration workflow

```
Term published
→ student receives registration window
→ student searches and compares sections
→ prerequisites/holds/capacity checked
→ student submits enrollment request
→ deterministic policy engine evaluates
→ enroll / waitlist / deny / approval required
→ audit event created
→ official record updated
→ student receives result
```

10. Student accounts and financial workflows

Screens

```
Student Finance
├── Account summary
├── Charges
├── Credits
├── Balance
├── Holds
├── Statements
├── Payment plans
├── Installments
├── Payment handoff
├── Refund status
├── Scholarship view
├── Aid handoff
├── Budget planner
├── Cost planner
└── Financial support resources

Staff Finance
├── Account queue
├── Charge/credit requests
├── Payment-plan review
├── Reconciliation
├── Month close
├── Dunning
├── Refund review
├── Disputes
├── Ledger audit
└── Reporting
```

11. Campus life and service super-app

Screens

```
Campus
├── Dining
├── Housing
├── Events
├── Organizations
├── Study spaces
├── Transit
├── Parking
├── Recreation
├── Library
├── Bookstore
├── Campus jobs
├── Volunteer work
├── Local services
├── Safety resources
├── Wellness resources
├── Accessibility resources
└── Emergency handoff
```

Dining workflow

```
Student opens Dining
→ sees location/hours/menu
→ views plan/balance where authorized
→ places order or receives official payment handoff
→ order state updates
→ ledger/reconciliation occurs
→ student receives status
```

12. Community and social connection

Screens

```
Community
├── Feed
├── Community directory
├── Community profile
├── Create post
├── Comments
├── Reactions
├── Membership
├── Alias settings
├── Events
├── Study sessions
├── Groups
├── Mentor discovery
```

```
├── Report content
├── Appeal
├── Guidelines
└── Settings

Moderation
├── Reports queue
├── Cases
├── Decisions
├── Appeals
├── Restrictions
├── Escalations
├── Volunteer tasks
├── Calibration
├── Safety entries
├── Policy
└── Audit
```

13. Career, employer, alumni, and lifelong learning

Screens

```
Career
├── Skills graph
├── Evidence
├── Verified claims
├── Credential wallet
├── Portfolio
├── Resume
├── Cover letter
├── Interview prep
├── Career plan
├── Opportunities
├── Applications
├── Employer profiles
├── Mentors
├── Alumni network
├── Career appointments
├── Career events
├── Continuing education
├── Microcredentials
└── Outcome history

Employer
├── Organization profile
├── Job/internship postings
├── Candidate criteria
├── Candidate search
├── Consent-based talent profiles
├── Interview scheduling
├── Hiring workflow
├── Internship management
└── Analytics
```

14. Family and guardian platform

Screens

```
Family Portal
├── Invitation
├── Relationship confirmation
├── Consent scope
├── Shared categories
├── Shared items
├── Time-bound access
├── Billing/support handoff
├── Resources
├── Communication preferences
├── Access history
├── Revoke access
└── Privacy information
```

Rule

```
Student grants access
→ family member accepts
→ only approved categories/resources visible
→ every access is auditable
→ access expires or is revoked
→ no broad academic surveillance
```

15. Institutional control plane

Screens

```
Institution Admin
├── Tenant overview
├── Campus hierarchy
├── Programs/departments
├── Identity providers
├── SSO
├── SCIM
├── Memberships
├── Roles
├── Capabilities
├── Access reviews
├── Tenant plan
├── Entitlements
├── Feature policies
├── Cohorts
├── Kill switches
├── Module modes
├── Rollout
├── Configuration Studio
```

```
├── Workflow Builder
├── Policy hierarchy
├── Governance stewards
├── Governance decisions
├── Integrations
├── Data classification
├── Consent policy
├── Retention policy
├── Legal holds
├── Data subject requests
├── AI provider policy
├── Approved learning sources
├── Compliance
├── Trust artifacts
├── Accessibility program
├── Incident notices
├── Risk register
├── Audit explorer
├── Offboarding
└── Export/portability
```

16. Integration, interoperability, and migration

Screens

```
Integration Hub
├── Provider directory
├── Connections
├── New connection
├── Security review
├── Scope approval
├── Mapping studio
├── Mapping versions
├── Field transforms
├── Sync schedule
├── Sync runs
├── Sync errors
├── Dead-letter queue
├── Replay requests
├── Reconciliation
├── Schema drift
├── Duplicate resolution
├── Source owners
├── Data freshness
├── Webhooks
├── LTI
├── OneRoster
├── Edu-API
├── Google/Microsoft
├── AI providers
└── Retirement
```

Integration workflow

```
Request
→ purpose
→ data classification
→ security/privacy review
→ scope approval
→ sandbox
→ mapping
→ read-only sync
→ reconciliation
→ controlled write request
→ approval
→ monitored production
→ periodic review
→ retirement/native replacement
```

17. Trust, security, privacy, and compliance

Public Trust Center

```
Trust Center
├── Security
├── Privacy
├── Accessibility
├── Responsible AI
├── Data practices
├── Subprocessors
├── Reliability/status
├── Compliance roadmap
├── HECVAT process
├── DPA process
├── Security document request
├── Trust room
└── Vulnerability report
```

Internal security and privacy

```
Security
├── Asset inventory
├── Access review
├── Secret inventory
├── Vulnerabilities
├── Dependencies
├── Threat models
├── Alerts
├── Incidents
├── Controls
├── Evidence
├── Vendor risk
├── Pen-test evidence
├── Break-glass review
```

```
├── Backup/restore
├── DR
└── Security reporting

Privacy
├── Data inventory
├── Data map
├── Classification
├── Retention
├── Deletion
├── Legal holds
├── Consent
├── Data requests
├── Export
├── Erasure
├── Subprocessors
├── Privacy incidents
└── Privacy reporting
```

18. Operations Command Center

Route map

```
/ops
├── Overview
├── Operations Inbox
├── My Work
├── Approvals
├── Tenants
├── Customers
├── Accounts
├── Pilots
├── Implementation
├── Success plans
├── Renewals
├── Customer health
├── Revenue
├── Billing
├── Support
├── Releases
├── Feature flags
├── Beta
├── Integrations
├── Trust
├── Compliance
├── Privacy
├── Security
├── Access reviews
├── Break-glass
├── Incidents
├── SLOs
├── Error budgets
├── Audit
├── Risks
```

```
├── Vendors
├── Finance
├── Forecasts
├── Reports
├── Board reports
├── Capacity
└── Preferences
```

Operating workflows

```
High-risk action:
request
→ evidence
→ approval
→ execution
→ audit
→ projection update
→ operator review

Support escalation:
ticket
→ severity
→ assignment
→ response
→ customer update
→ resolve
→ root cause
→ corrective action

Tenant rollout:
contract
→ implementation
→ evidence
→ launch gate
→ controlled launch
→ pilot
→ conversion
→ expansion/offboarding

Incident:
alert
→ triage
→ severity
→ command roles
→ mitigation
→ communication
→ recovery
→ postmortem
→ corrective work
```

19. Company operating system

Business systems

```
Company OS
├── Corporate governance
├── Legal operations
├── Finance
├── Budget
├── Runway
├── Pricing
├── Sales
├── Marketing
├── GTM
├── CRM
├── Pilots
├── Customer success
├── Support
├── Implementation
├── Billing
├── Collections
├── Partnerships
├── Vendor management
├── People/hiring
├── Product operations
├── Engineering
├── Security
├── Compliance
├── AI governance
├── Board/investor reporting
├── Risk
└── Operating cadence
```

Company workflows

```
Lead
→ qualification
→ discovery
→ demo
→ pilot design
→ security/procurement
→ contract
→ implementation
→ launch
→ adoption
→ pilot review
→ annual conversion
→ renewal
→ expansion
→ reference/case study
```

20. Developer platform and marketplace

Screens

```
Developer Portal
├── API documentation
├── API keys
├── OAuth clients
├── Service accounts
├── Webhooks
├── Event subscriptions
├── SDKs
├── Sandbox tenant
├── Test data
├── Integration certification
├── App marketplace
├── Partner directory
├── App review
├── Data scopes
├── Version management
├── Usage/metering
├── Support
└── Incident notices
```

21. Required document system

```
Product:
- Product thesis
- Architecture
- Role matrix
- Screen catalog
- Workflow catalog
- Data authority matrix
- Domain replacement matrix
- Release gates
- Roadmap
- Risk register

Academic:
- Course Studio
- Assessment
- Gradebook
- Registrar
- Academic records
- Degree audit
- Transfer
- Student success

AI:
- AI governance
- Provider registry
- Model routing
- Evaluation
```

```
- Red team
- Incident response
- Spend controls
- Course policy

Trust:
- Security
- Privacy
- Accessibility
- Data classification
- Retention
- Legal holds
- Incident response
- DR
- HECVAT
- Trust Center
- Vendor risk

Business:
- GTM
- Pricing
- Financial model
- Sales
- Pilot delivery
- Customer success
- Marketing
- Operations
- Board reporting
- Hiring plan

Engineering:
- Engineering standards
- Test strategy
- CI/CD
- SLOs
- Error budgets
- Observability
- CQRS/read models
- Operations Console
- Runbooks
- Backup/restore
```

22. Master traceability rule

Every screen, workflow, system, and document must be recorded with:

```
Name
Domain
Role
Route
Job to be done
Source authority
Data model
Tenant scope
Permission/capability
```

```
Consent requirement
AI policy
Accessibility requirement
Security requirement
Privacy/retention requirement
Audit event
Integration dependency
Migration dependency
Support owner
SLO/reliability class
Commercial relevance
Customer-success relevance
Required documentation
Test coverage
Rollback approach
Release gate
Current status
Evidence path
Owner
Priority
Dependency
Risk
Next action
```

Master execution command

```
Create the definitive Semester Master Catalog.

Semester is the unified AI-native operating system for education. It includes
student productivity, learning, LMS, SIS/registrar, advising, student success,
financial workflows, campus services, community, career, family/guardian,
institutional governance, integrations, security, privacy, AI, marketplace,
company operations, and developer platform capabilities.

Audit the current Semester repository and database first.

Then create:

docs/master/SEMESTER_COMPLETE_CATALOG.md
docs/master/SEMESTER_SCREEN_CATALOG.md
docs/master/SEMESTER_ROLE_CATALOG.md
docs/master/SEMESTER_WORKFLOW_CATALOG.md
docs/master/SEMESTER_SYSTEM_CATALOG.md
docs/master/SEMESTER_CONSOLE_CATALOG.md
docs/master/SEMESTER_DATA_DOMAIN_CATALOG.md
docs/master/SEMESTER_EDUCATION_GRAPH.md
docs/master/SEMESTER_DOCUMENT_CATALOG.md
docs/master/SEMESTER_INTEGRATION_CATALOG.md
docs/master/SEMESTER_RELEASE_GATE_CATALOG.md
docs/master/SEMESTER_TRACEABILITY_MATRIX.md
docs/master/SEMESTER_GAP_AND_STATUS_REGISTER.md

For every catalog item include:
- domain
- role
```

```
- screen/route
- workflow
- source authority
- data classification
- tenant scope
- permission/capability
- consent requirement
- AI policy
- accessibility requirement
- security/privacy requirement
- audit requirement
- integration/migration dependency
- support owner
- SLO
- tests
- release gate
- current verified status
- evidence
- priority
- risk
- next action

Do not mark an item complete based only on a screen, document, table, or
migration. It is complete only when code, authorization, tenant isolation,
accessibility, tests, monitoring, support, documentation, and release evidence
exist.

Return:
- every role
- every screen
- every workflow
- every system
- every console
- every document
- every graph/domain
- every integration
- every release gate
- every P0/P1/P2/P3 gap
- a branch-by-branch implementation plan
- the top 50 actions in priority order.
```

The most important organizing principle is simple:

> **Students should experience one coherent system. Institutions should operate one governed platform. Semester should run one accountable company.**

⁂

1. expand-evan-further-provide-prompt-and-comand-to-e.pdf

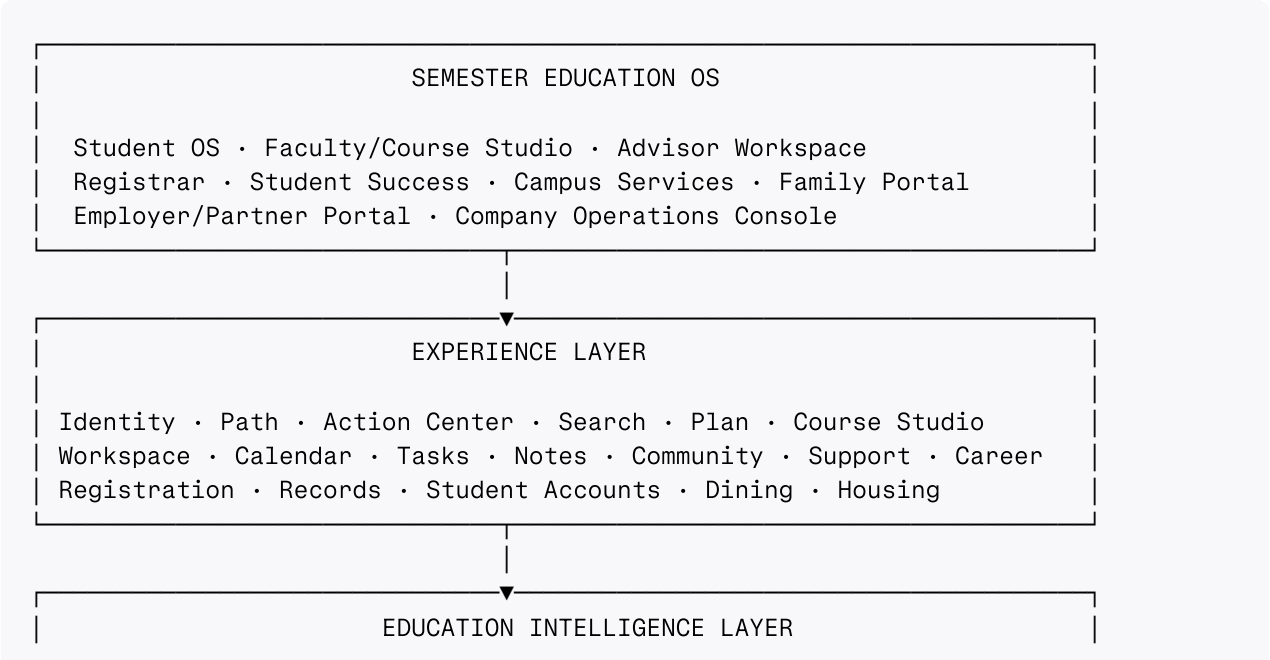
Central Education OS architecture map: how Semester maps to Canvas, SIS registrars, Brightspace, and AI models like Claude and ChatGPT

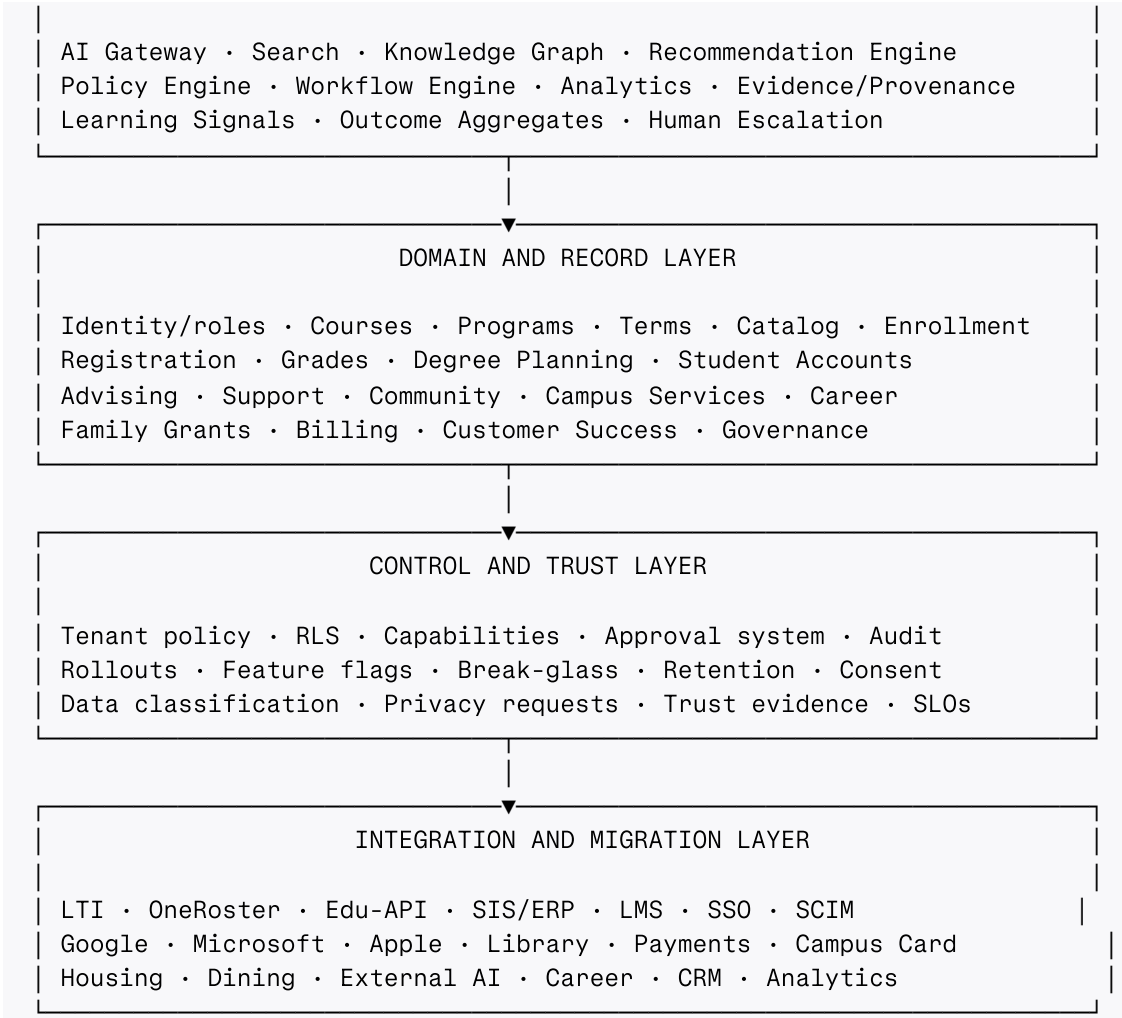
Computer
Semester vs legacy campus tech: feature-by-feature integration matrix comparing SIS, LMS, Google/Microsoft, and AI services
Computer
Map out the integration and migration strategy for LTI and OneRoster
Draft the technical architecture for the multi-tenant institutional OS
Build the student-facing daily productivity and AI copilot interface

Semester should be designed as a **unified Education OS** with one education graph, one policy/control plane, and multiple role-specific experiences. It can connect to Canvas, Brightspace, SIS/registrar platforms, Google/Microsoft productivity tools, and AI providers during transition—then become the native authoritative system for individual domains as institutions approve migration.

The practical architecture is: **connect first, replace by domain, operate as one system.** LTI supports secure learning-tool connections to institutional learning environments, while OneRoster supports standardized exchange of users, courses, enrollments, resources, and grades across education systems. [1] [2] [3]

Central Education OS map





Semester vs legacy systems

Legacy category	Typical tools	Current limitation	Semester integration phase	Semester end-state
LMS/course platform	Canvas, Brightspace, Blackboard, Top Hat	Course experience is separated from planning, support, career, and daily work	LTI launch, roster/course sync, assignment/resource import	Native Course Studio, assignments, study packs, feedback, grade workflows, governed AI
SIS/registrar	Banner, PeopleSoft, Workday Student, Colleague, Jenzabar	Records/registration are disconnected from student guidance and action	Read-only roster/catalog/term/hold sync, official handoffs	Native enrollment, catalog, prerequisites, holds, overrides, registration, academic record workflows

Legacy category	Typical tools	Current limitation	Semester integration phase	Semester end-state
Degree audit/advising	Degree Works, Stellic, advising CRMs	Planning is separate from daily student work and institutional support	Import program requirements and official progress	Semester Path, scenarios, degree planning, advisor workflows, source-aware requirement checks
Student success	EAB Navigate, Starfish, Civitas-style tools	Alerts, advising, engagement, and interventions are isolated	Import authorized signals and referrals	Native Action Center, support workflow, case management, outcome intelligence
Productivity	Google Workspace, Microsoft 365, Apple Notes/Calendar	Generic tools lack academic context	Calendar/file/account integration	Native education workspace: tasks, plans, notes, project work, study sessions, AI
AI tools	ChatGPT, Claude, Perplexity, Copilot	No institutional policy, course context, source authority, cost controls, or audit	Approved-model routing via AI gateway	Semester AI copilot with policy, evidence, provenance, course rules, human escalation
Search	Google Search, campus site search, library discovery	Student must know where to search; results lack role/policy context	Federated search/indexing	Permission-aware education search with official sources, citations, context, and action
Campus services	Separate housing, dining, transit, event, safety apps	Campus life is fragmented from student plan	Read-only feeds and approved action handoffs	Native service action layer where institution chooses it
Billing/student accounts	SIS finance modules, payment portals	Financial obligations are opaque and disconnected from deadlines/support	Account-balance/status integration	Native student-account ledger, payment plans, reconciliation, official payment-provider workflows
Career	Handshake, LinkedIn, career portals	Skills, coursework, projects, advising, and opportunity discovery are disconnected	Opportunity import and approved profile sharing	Career graph, verified skills, portfolio, mentoring, opportunities, employer relationships

Legacy category	Typical tools	Current limitation	Semester integration phase	Semester end-state
Family/guardian	Separate portals and email workflows	Sharing is often overly broad or opaque	Identity-linking and consent-aware handoffs	Student-controlled grants, time-bound sharing, audit history, scoped support
Company operations	CRM, support desk, billing, spreadsheet stacks	Product, revenue, support, pilots, trust, and rollout data remain siloed	Read-model integration	Semester Operations Command Center

Integration strategy

Principles

1. **Do not break institutional authority.** Existing official records remain authoritative until a formal domain migration is approved.
2. **Minimum data first.** Pull only the fields needed for the workflow.
3. **Read before write.** Begin with read-only sync; enable writes only after reconciliation, governance, rollback, and approval.
4. **Every integration has a purpose.** No “connect everything” scope.
5. **Every field has provenance.** Source, freshness, transformation, owner, and conflict state must be visible.
6. **Every write has an idempotency key.** Retrying must never create duplicate enrollment, payment, grade, or record changes.
7. **Every migration has rollback.** Dual-run, reconciliation, cutover, and restoration must be explicit.
8. **Every external model is policy-governed.** AI providers receive only allowed data under tenant/course policy.

Integration lifecycle

```
Inventory
  ↓
Business purpose
  ↓
Data classification
  ↓
Security/privacy review
  ↓
Scope approval
  ↓
Sandbox/test connection
  ↓
Read-only sync
```

```
  ↓
Data mapping and reconciliation
  ↓
Pilot workflow
  ↓
Controlled write approval
  ↓
Production monitoring
  ↓
Periodic access and scope review
  ↓
Retirement or native replacement
```

Integration connection record

```
type IntegrationConnection = {
  id: string;
  tenantId: string;
  provider: string;
  systemType:
    | "sis"
    | "lms"
    | "identity"
    | "productivity"
    | "ai"
    | "library"
    | "payments"
    | "campus-service"
    | "career"
    | "analytics";

  status:
    | "draft"
    | "security_review"
    | "approved"
    | "sandbox"
    | "read_only"
    | "write_enabled"
    | "paused"
    | "retired";

  direction: "read" | "write" | "bidirectional";
  approvedScopes: string[];
  dataClassification: string[];
  ownerRole: string;

  lastSuccessfulSyncAt?: string;
  freshnessTargetMinutes?: number;
  reconciliationStatus:
    | "not_started"
    | "passing"
    | "warning"
    | "failed";

  rollbackPlan: string;
```

```
  approvalReference: string;
};
```

LTI strategy

LTI should be the first LMS connection path where Semester behaves as a tool launched from Canvas, Brightspace, Blackboard, or another compatible learning environment.

LTI use cases

LTI workflow	Semester experience
Course launch	Student opens Semester from course context without a separate login flow
Course membership	Semester receives authorized context about the course, user, and role
Study workspace	Student enters course-specific notes, tasks, study packs, and AI support
Instructor tools	Faculty publish course guidance, AI rules, approved sources, study packs
Assignment support	Semester links action plans and preparation to course tasks
Grade passback	Only after explicit institutional approval and controlled grade workflow
Analytics	Aggregate course engagement/workflow signals subject to tenant policy

LTI is a standard for connecting learning tools to institutional learning environments, helping tools receive context such as roles, enrollment, and course context without separate sign-in flows.[1]

LTI implementation sequence

```
1. Register Semester as an LTI tool.
2. Validate OIDC login, state, nonce, issuer, audience, deployment, and signatures.
3. Bind LTI launch identity to a Semester account through controlled linking.
4. Create or refresh tenant/course membership only from approved source claims.
5. Open course-scoped Semester experience.
6. Store no unnecessary raw launch claims.
7. Audit launch validation and account-linking events.
8. Support deep links to course-specific views.
9. Begin read-only.
10. Enable grade passback only after:
    - institution approval
    - role/capability verification
    - grade policy mapping
    - idempotency
    - reconciliation
    - audit
    - rollback process
```

OneRoster strategy

OneRoster should handle structured exchange of users, organizations, courses, classes, enrollments, resources, and grades where supported.

OneRoster is designed to exchange roster, course, learning-resource, and grade information between education systems; it supports both CSV and REST-based models.[2] [3]

OneRoster migration map

OneRoster entity	Semester canonical model	Migration use
Org	Institution, campus, school, department	Tenant hierarchy
User	Person, profile, membership	Provisioning and identity linkage
Academic session	Term	Term calendar and policy scope
Course	Catalog course	Official course reference
Class	Section/course offering	Course instance and schedule
Enrollment	Institution membership/course enrollment	Access and course context
Resource	Course material/study pack reference	Approved source/resource links
Result/grade	Grade ledger/version	Controlled read/import or passback
Line item	Gradebook item	Assessment mapping

OneRoster migration phases

```
Phase A — CSV import
- Validate schema.
- Map external identifiers.
- Stage rows.
- Report duplicates/errors.
- Reconcile counts.
- Keep source authority external.

Phase B — OneRoster REST read sync
- Incremental roster/course/term sync.
- Per-record freshness.
- Soft deletes.
- Source snapshots.
- Reconciliation dashboards.

Phase C — Controlled workflow exchange
- Resource publishing.
- Course-context sync.
- Approved grade/line-item exchange.
- Strict scope and audit controls.

Phase D — Native Semester domain
- Semester becomes authoritative for approved records.
- Dual-run with old system.
```

```
- Migration evidence.
- Reconciliation.
- Institutional acceptance.
- Cutover.
- Rollback window.
```

Multi-tenant institutional OS architecture

Tenant hierarchy

```
Semester Platform
└── System
    └── Institution
        └── Campus
            └── School/College
                └── Department
                    └── Program
                        └── Course
                            └── Section
                                └── Cohort
                                    └── Person
```

Policy narrows as it moves downward:

```
Platform policy
  ↓
Institution policy
  ↓
Campus policy
  ↓
School policy
  ↓
Program policy
  ↓
Course policy
  ↓
Individual consent and grants
```

A lower policy can become stricter but should not silently override a higher-risk platform/institution restriction.

Multi-tenant access model

```
Identity
  ↓
Authenticated account
  ↓
Institution membership
  ↓
Role grant
```

```
  ↓
Capability
  ↓
Scope:
- global
- tenant
- campus
- school
- program
- course
- section
- individual
  ↓
Purpose
  ↓
Time limit
  ↓
Audit
```

Every sensitive decision must answer:

```
Who is acting?
What capability do they have?
What tenant/campus/course scope applies?
Why is this access needed?
What data category is involved?
How long does the grant last?
What audit event is recorded?
Can the action be reversed?
```

Core institutional OS domains

```
Identity and provisioning
├── SSO
├── SCIM
├── Membership
├── Roles
├── Capabilities
└── Access review

Academic operations
├── Terms
├── Catalog
├── Sections
├── Enrollment
├── Registration
├── Holds
├── Prerequisites
├── Overrides
├── Grades
├── Academic records
├── Degree planning
├── Transfer evaluation
```

```
└── Graduation

Learning
├── Course Studio
├── Assignments
├── Study packs
├── Course guidance
├── AI policy
├── Gradebook
├── Feedback
├── Integrity
└── LTI

Student success
├── Action Center
├── Advising
├── Support
├── Early signals
├── Referrals
├── Check-ins
├── Office actions
└── Outcome analysis

Campus life
├── Dining
├── Housing
├── Events
├── Organizations
├── Community
├── Safety
├── Wellness handoffs
└── Campus services

Commerce and operations
├── Billing
├── Student accounts
├── Payment plans
├── Financial aid handoffs
├── Contracts
├── Commercial plans
├── Entitlements
└── Customer success

Trust and governance
├── Privacy
├── Consent
├── Retention
├── Legal holds
├── Audit
├── Human overrides
├── Compliance
├── Trust room
├── Risk
└── Incident response
```

Student-facing daily productivity and AI copilot

The student experience should not resemble a dashboard full of widgets. It should act as a daily **decision and action environment**.

Daily workspace

```
┌──────────────────────────────────────────────────────────┐
| Semester                                                 |
| Good evening, Harrison.                                  |
|                                                          |
| Today                                                    |
| ├── 2 urgent actions                                     |
| ├── 1 upcoming registration deadline                     |
| ├── 3 study tasks                                        |
| └── 1 support/resource recommendation                    |
|                                                          |
| Your path                                                |
| ├── Current term progress                                |
| ├── Degree / goal progress                               |
| ├── Schedule and workload                                |
| └── Next milestone                                       |
|                                                          |
| Courses                                                  |
| ├── Course Studio shortcuts                              |
| ├── Upcoming work                                        |
| ├── Study packs                                          |
| └── Course AI guidance                                   |
|                                                          |
| Copilot                                                  |
| “What do you need to understand, decide, or do next?”    |
|                                                          |
| Action Center                                            |
| ├── Registration                                         |
| ├── Advising                                             |
| ├── Financial / account notices                          |
| ├── Campus services                                      |
| └── Career opportunities                                 |
└──────────────────────────────────────────────────────────┘
```

Core student navigation

```
Home
Path
Plan
Courses
Workspace
Calendar
Actions
Search
Community
Support
```

```
Career
Profile
```

AI copilot architecture

```
Student request
  ↓
Intent classification:
- explain
- summarize
- plan
- compare
- draft
- research
- study
- navigate
- troubleshoot
- escalate
  ↓
Context selection:
- course
- term
- plan
- calendar
- institution policy
- approved source
- student-owned workspace
  ↓
Policy check:
- role
- tenant
- course AI rules
- data classification
- consent
- provider restriction
- integrity rules
- budget
  ↓
Response:
- answer
- source citations
- confidence/limits
- action suggestions
- official handoff
- human escalation
  ↓
Evidence:
- source references
- policy applied
- action log
- user feedback
```

Copilot interaction principles

Principle	Product behavior
Explainable	“Here is why this recommendation appears.”
Source-aware	Official records, course content, approved sources, and public research are clearly labeled
Non-authoritative by default	AI guidance does not silently become an official institutional decision
Action-linked	The copilot can create draft tasks, plans, notes, or navigate to official handoffs
Permission-bounded	It only sees what the user/tenant/policy permits
Human-aware	It escalates to advisor, instructor, registrar, support, or office when appropriate
Integrity-aware	Course-level AI guidance constrains study, drafting, assessment, and citation behavior
Cost-aware	Shows or meters expensive AI actions and respects tenant budget
Student-controlled	Student can inspect, correct, disable, or limit personal AI memory where appropriate

Copilot example

```
Student:
“Can I register for BIO 221 next term?”

Semester:
“You appear to have completed BIO 101, which is listed as a prerequisite
for BIO 221 in the current catalog record.

Before you register:
- Your registration window opens on [date/time].
- There is one advising hold listed by [office].
- BIO 221 has [available/waitlist] status.
- Your current plan would total [X] credits.

This is planning guidance, not an official enrollment decision.
Open Registration → Review the official section and hold details.”
```

This response:

- Uses authoritative-source labels.
- Explains data source and limits.
- Does not make an unsupported official decision.
- Links to the correct action.
- Helps the student act without hiding institutional authority.

Build prompt

```
Build the Semester Central Education OS architecture and role-specific
experiences.

Semester's product thesis is to become the unified operating system for
education: student OS, productivity suite, course/learning platform,
institutional OS, AI platform, campus super-app, registrar/SIS system,
marketplace, family/guardian platform, career platform, and company operations
platform.

The strategy is:

CONNECT FIRST.
REPLACE BY DOMAIN.
OPERATE AS ONE SYSTEM.

Build the system as one coherent education graph, not disconnected modules.

Implement or document:

1. Central Education OS architecture map
2. Legacy-system integration/replacement matrix:
   - Canvas
   - Brightspace
   - Blackboard
   - Top Hat
   - SIS/ERP/Registrar
   - degree audit
   - advising/student success
   - Google
   - Microsoft
   - Apple productivity
   - ChatGPT
   - Claude
   - Perplexity
   - campus services
   - career platforms
3. LTI integration strategy
4. OneRoster import/sync/migration strategy
5. Edu-API/interoperability strategy where applicable
6. Multi-tenant institutional hierarchy
7. Role/capability/scope model
8. Source-of-authority matrix
9. Connect-to-native replacement roadmap
10. Student daily workspace
11. Student AI copilot
12. Course Studio
13. Academic/registrar architecture
14. Campus services architecture
15. Family/guardian consent/grant model
16. Career and lifelong-learning architecture
17. Institutional control plane
18. AI governance/model-routing layer
19. Migration, reconciliation, dual-run, cutover, rollback, and offboarding
20. Security, privacy, audit, retention, accessibility, and support requirements
```

```
For every domain define:
- user roles
- jobs to be done
- authoritative source
- tenant boundary
- permission model
- data classification
- integration path
- native replacement state
- migration plan
- audit trail
- rollback
- support model
- success metrics
- launch gates
- commercial model

Do not claim an existing native capability unless code, migration, test, and
operational evidence prove it exists.

Do not replace official records, grades, registration, payment, identity, or
legal processes without formal institutional authority, tested migration,
reconciliation, auditability, rollback, and approval.

The student experience must feel unified:
Identity → Path → Plan → Courses → Actions → Support → Career.

The final result must make users feel:
“This is not another portal. This is where my educational life works.”
```

⁂

1. https://www.1edtech.org/standards/lti
2. https://www.1edtech.org/standards/oneroster
3. https://www.1edtech.org/blog/rostering-resources-and-gradebook-standards

perplexity

expand evan further provide prompt and comand to excute

Semester's ambition should be expanded beyond "unifying existing tools" into a complete doctrine for becoming the **foundational education platform**: the common operating layer for learning, academic progress, identity, records, campus services, productivity, AI, institutional operations, and lifelong outcomes.

The key is to preserve the full replacement thesis while making it technically and institutionally credible:

> **Build every major domain natively. Integrate during transition. Replace only after evidence. Operate every domain as one coherent system.**

Semester can be comprehensive from inception without pretending every legacy migration is complete on day one. The platform must have native modules for every major domain, while integration, dual-run, reconciliation, accessibility, migration, institutional approval, support, and rollback determine when it becomes authoritative for each domain. Standards like LTI, OneRoster, and Edu-API provide transition mechanisms for learning-tool context, roster/course/grade exchange, and academic enterprise interoperability. [1] [2] [3]

Expanded Semester doctrine

The mission

> **Semester is the operating system for education: a unified, AI-native platform that brings the entire educational life cycle—identity, learning, planning, records, services, community, career, administration, intelligence, and operations—into one trusted system.**

The long-term goal is not a portal that aggregates links. It is not a dashboard layered on top of a dozen systems. It is not merely an LMS, SIS, advisor tool, or AI assistant.

Semester is intended to become:

```
The student's daily operating system
The educator's teaching and learning workspace
The advisor's action and support environment
The registrar's academic workflow engine
The institution's governance and intelligence layer
The family's consent-based support channel
```

```
The employer’s verified talent pathway
The company’s commercial and operating platform
```

The full replacement thesis

```
Fragmented education stack
  ↓
Semester connects the pieces
  ↓
Semester centralizes the experience
  ↓
Semester owns the workflow layer
  ↓
Semester becomes native system of record by domain
  ↓
Semester becomes the central education operating system
```

The intended endpoint:

```
Before Semester:
Student uses:
- LMS
- SIS portal
- registrar portal
- degree audit
- Google Drive
- Microsoft Office
- calendar app
- note app
- task app
- ChatGPT
- Perplexity
- advising portal
- tutoring portal
- library system
- career portal
- student account portal
- housing portal
- dining app
- events app
- campus safety app
- group-chat tools
- support ticket portal

After Semester:
Student uses:
Semester
```

Behind the scenes, Semester either:

1. **Natively operates the domain**
2. **Synchronizes a transitional external domain**

3. **Uses a specialist partner only where required**
4. **Provides an official handoff until institutional authorization permits replacement**

The student should not have to know which architecture is active. The user sees one coherent path.

The complete product universe

1. Personal education OS

```
Semester Home
├── Today
├── Action Center
├── Calendar
├── Tasks
├── Notes
├── Files
├── Plans
├── Goals
├── Reminders
├── Focus
├── Time and workload
├── Personal AI copilot
├── Search
├── Notifications
├── Profile and settings
└── Cross-device sync
```

This replaces the student's fragmented use of calendar apps, task apps, notes apps, reminders, generic cloud folders, generic productivity suites, and disconnected AI chats.

2. Academic path OS

```
Academic Path
├── Institution
├── Program
├── Degree requirements
├── Course history
├── Current courses
├── Future terms
├── Prerequisites
├── Registration eligibility
├── Holds
├── Transfer credit
├── Articulation rules
├── Degree audit
├── Graduation scenarios
├── Major/minor comparison
├── Workload planning
├── Cost projections
```

```
├── Credential milestones
└── Official record handoffs
```

This replaces the fragmented experience across degree-audit systems, PDF catalogs, registrar portals, spreadsheets, advising notes, email chains, and unofficial student planning tools.

3. Learning and course OS

```
Course Studio
├── Course home
├── Objectives
├── Syllabus
├── Schedule
├── Announcements
├── Assignments
├── Assessment calendar
├── Resources
├── Reading and annotation
├── Notes
├── Study packs
├── Flashcards
├── Practice
├── Tutor workflows
├── Discussions
├── Groups
├── Office hours
├── Feedback
├── Gradebook
├── Regrade process
├── Academic integrity guidance
├── Course AI rules
├── Research sources
└── Learning evidence
```

This replaces or absorbs functions typically scattered across Canvas, Brightspace, Blackboard, Top Hat, publisher courseware, study apps, flashcard apps, generic note apps, email, group chat, and isolated AI tools.

4. AI-native education OS

```
Semester AI
├── Copilot
├── Course-aware tutor
├── Research assistant
├── Writing assistant
├── Study planner
├── Registration guide
├── Academic path explainer
├── Advisor support assistant
├── Faculty content assistant
├── Staff workflow assistant
├── Institutional knowledge assistant
```

```
├── Search and source synthesis
├── Agentic workflow execution
├── Model routing
├── Policy engine
├── Data classification
├── Consent controls
├── Source provenance
├── Cost controls
├── Integrity controls
├── Human escalation
└── AI evaluation and audit
```

Semester's goal is not to build one monolithic model that defeats every foundation model. It is to become the **trusted education intelligence layer** that makes any approved model useful, safe, contextual, source-aware, policy-bound, auditable, and action-connected.

```
ChatGPT / Claude / Perplexity / other models
                 ↓
          Semester AI Gateway
                 ↓
          Education context
                 ↓
Course policy + institutional policy + consent + permissions
                 ↓
Approved sources + student workspace + official records
                 ↓
Explainable answer + safe action + audit + human handoff
```

NIST's AI Risk Management Framework provides a useful reference for designing, deploying, governing, and evaluating AI systems with explicit risk management and trustworthiness considerations.[4] [5]

5. Advisor and student-success OS

```
Student Success Workspace
├── Caseload
├── Student path view
├── Action completion
├── Support signals
├── Check-ins
├── Appointments
├── Outreach
├── Referrals
├── Student communication
├── Early support indicators
├── Intervention workflows
├── Success plans
├── Progress reviews
├── Outcome analytics
├── Escalation rules
├── Privacy/consent scope
└── Aggregate institutional insights
```

This replaces disconnected advising CRMs, alert systems, spreadsheets, email workflows, appointment tools, support ticket systems, and generic retention dashboards.

6. Registrar and academic operations OS

```
Registrar OS
├── Academic calendar
├── Terms
├── Course catalog
├── Sections
├── Meeting times
├── Capacity
├── Waitlists
├── Prerequisites
├── Registration windows
├── Time tickets
├── Holds
├── Overrides
├── Enrollment
├── Add/drop/withdrawal
├── Academic record ledger
├── Grade ledger
├── Grade changes
├── Regrades
├── Graduation checks
├── Transfer credit
├── Academic policy rules
├── Audit reports
└── Official export/integration channels
```

The full vision includes native registrar capability, but it must only become authoritative through formal institutional approval, migration evidence, reconciliation, tested rollback, policy review, and controlled cutover.

7. Student finance and account OS

```
Student Financial OS
├── Student account
├── Charges
├── Credits
├── Balance
├── Payment plan
├── Due dates
├── Holds
├── Reconciliation
├── Refund status
├── Financial-aid handoffs
├── Scholarships
├── Budget tools
├── Cost planning
├── Institutional payment provider
```

```
├── Family consent-based support
└── Audit ledger
```

This replaces the fragmented student experience around bursar portals, payment-plan tools, spreadsheet budgets, emails, billing notices, and opaque account holds.

8. Campus life OS

```
Campus OS
├── Housing
├── Dining
├── Campus card
├── Events
├── Student organizations
├── Community
├── Study spaces
├── Transit
├── Campus safety
├── Wellness handoffs
├── Accessibility services
├── Recreation
├── Library
├── Bookstore
├── Campus jobs
├── Volunteer work
├── Local services
└── Emergency communications
```

The platform should not automatically centralize sensitive health, safety, conduct, or emergency data. It should use purpose limitation, strict tenant policy, explicit authority, and safe official handoffs.

9. Career and lifelong learner OS

```
Career OS
├── Skills graph
├── Evidence portfolio
├── Resume
├── Project history
├── Credentials
├── Internship search
├── Jobs
├── Employer profiles
├── Alumni mentors
├── Career appointments
├── Interview preparation
├── Opportunity matching
├── Verified learning evidence
├── Continuing education
├── Credential renewal
└── Alumni relationship
```

Semester's opportunity is to connect learning evidence to career outcomes without turning student data into ungoverned surveillance or a data-broker product.

10. Family and guardian OS

```
Family Support OS
├── Consent-based family invitations
├── Scoped permissions
├── Time-bounded access
├── Chosen categories
├── Resource sharing
├── Billing/support handoffs
├── Progress summaries
├── Communication preferences
├── Revocation
├── Audit history
└── No broad academic surveillance
```

The student must control sharing. No "parent portal" should expose more than what the student and institution authorize.

11. Faculty and academic leadership OS

```
Faculty Workspace
├── Course Studio
├── Syllabus and policy publishing
├── Course AI rules
├── Resources
├── Assessments
├── Gradebook
├── Feedback
├── Regrade workflow
├── Office hours
├── Teaching assistants
├── Learning signals
├── Course improvement
├── Curriculum alignment
├── Accessibility checks
├── Academic-integrity workflows
└── Cross-course outcomes
```

12. Institutional governance OS

```
Institutional Control Plane
├── Identity and provisioning
├── Roles and capabilities
├── Tenant policy
├── Feature controls
├── Configuration studio
├── Workflow builder
├── Integration governance
```

```
├── Data classification
├── Retention
├── Consent
├── Legal holds
├── Privacy requests
├── Accessibility governance
├── AI provider policy
├── Source authority
├── Rollout controls
├── Security controls
├── Audit
├── Incident response
├── Compliance evidence
├── Trust room
├── Procurement
└── Offboarding and portability
```

13. Developer and platform ecosystem

```
Semester Platform Ecosystem
├── Developer portal
├── APIs
├── Webhooks
├── OAuth/OIDC
├── LTI
├── OneRoster
├── Edu-API
├── SCIM
├── App marketplace
├── Partner directory
├── Integration certification
├── Sandbox environments
├── SDKs
├── Event subscriptions
├── Extension framework
├── Partner governance
└── Revenue sharing
```

This is how Semester becomes a platform rather than a closed product.

Deeper strategic moat

1. Education graph moat

The moat is not one feature. It is the relationship between all the data and workflows:

```
Student
→ course
→ assignment
→ skill
→ academic requirement
→ task
```

```
→ resource
→ advisor
→ support office
→ institutional policy
→ career opportunity
→ verified outcome
```

Once Semester is the system where these relationships live, competitors must rebuild the graph, policy engine, integrations, workflow history, and trust evidence—not merely match a screen.

2. Policy and trust moat

```
AI rule
+ course rule
+ institution rule
+ consent
+ data classification
+ role
+ tenant scope
+ source authority
+ audit
+ human override
= trusted educational action
```

Generic AI companies cannot easily offer this without becoming education infrastructure. Legacy campus vendors cannot easily offer it without rebuilding their architecture around modern AI, student experience, cross-domain workflows, and policy-aware intelligence.

3. Migration moat

Semester should make migration a product:

```
Inventory
→ map
→ import
→ reconcile
→ dual run
→ validate
→ migrate users
→ cut over
→ monitor
→ rollback if needed
→ retain/export/archive
```

Every migration creates institutional knowledge, field maps, quality rules, reconciliation evidence, and implementation expertise that improve the next migration.

4. Outcome moat

Semester must prove:

```
Student:
- understands next steps faster
- completes more critical actions
- experiences less administrative friction
- gains clearer path and career visibility

Institution:
- sees fewer avoidable support contacts
- gains better action completion
- improves implementation consistency
- receives explainable aggregate insights
- reduces fragmented vendor burden

Faculty/staff:
- spend less time chasing information
- use clearer workflows
- have safer, more governed AI support
- retain human judgment where it matters
```

Complete domain replacement matrix

Domain	Transition integration	Native Semester capability	Replacement gate
Identity	SSO, SCIM, directory sync	Identity membership, role grants, capabilities	Provisioning reconciliation, access review, rollback
LMS	LTI, roster/course read sync	Course Studio, study packs, learning workflows	Faculty acceptance, accessibility, grade workflow parity
SIS	Read-only APIs, OneRoster/Edu-API, imports	Canonical academic data and policy workflows	Data reconciliation, legal/registrar approval, audit
Registration	SIS handoff	Native terms, sections, enrollment, waits, holds, overrides	Transaction integrity, capacity correctness, rollback
Gradebook	LMS passback/read	Native grade ledger, assessments, regrades	Faculty policy, student visibility, official-record governance
Degree audit	Requirement import	Path engine, scenarios, requirement evaluation	Registrar validation and official-signoff workflow
Advising	CRM/referral sync	Caseload, success plan, action center, referrals	Privacy, caseload, intervention, outcome validation
Student accounts	Balance/status sync	Ledger, requests, payment plans, reconciliations	Finance controls, payment reconciliation, audit
Productivity	Google/Microsoft/Apple integrations	Notes, tasks, calendar, files, projects	Data portability, reliability, user acceptance
AI	Approved model routing	Copilot, tutor, research, agent workflows	Policy, evaluation, course rules, cost control

Domain	Transition integration	Native Semester capability	Replacement gate
Campus services	Feed/handoff	Native service workflows where approved	Service-owner approval, safety/privacy review
Career	Opportunity import	Skills graph, portfolio, matching	Employer data rules, student consent, outcome evidence
Family	Existing communication handoffs	Scoped grants, sharing, audits	Student consent, institution policy, revocation
Support	Existing helpdesk integration	Support tickets, help requests, knowledge base	SLA, escalation, privacy, operational readiness
Company operations	CRM/billing/support integrations	Operations Command Center	Internal controls, finance/security review

Semester build order

The thesis is full replacement from inception. The **build order** must still respect dependency and risk:

```
Foundation:
Identity, tenancy, RLS, audit, policy, data classification, consent,
integrations, rollout, support, observability.

Daily student OS:
Home, Action Center, Path, Plan, Calendar, Tasks, Notes, Search, Copilot.

Learning:
Course Studio, resources, study, course policies, AI rules, learning evidence.

Institutional workflow:
Advising, registration readiness, tenant configuration, implementation, pilots.

Core academic operations:
Catalog, sections, registration, gradebook, records, degree progression.

Campus life and services:
Dining, events, community, organizations, spaces, support, accessibility.

Career and lifelong relationship:
Skills, portfolio, mentors, opportunities, employer ecosystem, alumni.

Platform:
Developer ecosystem, marketplace, APIs, extensions, multi-campus governance.

Company command center:
GTM, billing, customer success, trust, compliance, finance, security,
operations, SLOs, releases, support, renewals.
```

Expanded master execution prompt

Use this with Claude Code, Codex, or a GitHub-connected coding agent.

```
You are the combined Chief Product Officer, Chief Technology Officer,
Chief Learning Officer, Chief Academic Officer, Chief Information Security
Officer, Chief Operating Officer, Head of Student Success, Registrar Systems
Architect, AI Governance Lead, and Platform Ecosystem Architect for Semester.

Repository:
https://github.com/harrisonjrubin7-cmyk/semester

Semester's thesis is comprehensive:

Semester is the unified, AI-native operating system for education.

It is designed from inception to become:

- Student operating system
- Productivity suite
- Course and learning platform
- LMS replacement
- SIS and registrar replacement
- Academic record and degree-planning system
- Advising and student-success system
- Student financial/account system
- Campus services super-app
- AI copilot and governed intelligence platform
- Search and knowledge platform
- Family/guardian consent platform
- Career and lifelong learning platform
- Institutional control plane
- Developer ecosystem and marketplace
- SaaS company operating platform

The long-term purpose is to replace the fragmented technology stack currently
spread across:

- ChatGPT
- Claude
- Perplexity
- generic AI assistants
- Google Workspace
- Microsoft 365
- Apple productivity workflows
- Canvas
- Brightspace
- Blackboard
- Top Hat
- publisher courseware
- SIS/ERP
- Registrar
- degree audit
- academic records
- registration systems
- gradebooks
```

```
- advising CRMs
- student-success platforms
- student account portals
- financial aid handoffs
- housing portals
- dining apps
- campus safety apps
- community platforms
- career systems
- support portals
- family portals
- separate campus services
- disconnected company operations tools

The strategy is:

CONNECT FIRST.
REPLACE BY DOMAIN.
OPERATE AS ONE SYSTEM.

Do not narrow Semester's product thesis.

However:

- Do not falsely claim native replacement where only an integration exists.
- Do not replace official records, grades, registration, financial records,
  identity, or legal processes without formal authority, migration, audit,
  reconciliation, rollback, and institution approval.
- Do not create disconnected feature silos.
- Do not treat generic AI output as official institutional truth.
- Do not compromise tenant isolation, student agency, accessibility, privacy,
  or source authority.

Build a complete architecture and execution program.

============================================================
PHASE 0 — UNIFIED PRODUCT AUDIT
============================================================

Read all relevant code, migrations, architecture, feature inventory, audit,
security, governance, integration, product, and operations documentation.

Create:

docs/product/SEMESTER_UNIFIED_EDUCATION_OS_THESIS.md
docs/product/EDUCATION_OS_PRODUCT_MAP.md
docs/product/EDUCATION_GRAPH_ARCHITECTURE.md
docs/product/ROLE_EXPERIENCE_MATRIX.md
docs/product/DOMAIN_AUTHORITY_MATRIX.md
docs/product/DOMAIN_REPLACEMENT_MATRIX.md
docs/product/CONNECT_REPLACE_OPERATE_STRATEGY.md
docs/product/INTEROPERABILITY_AND_MIGRATION_STRATEGY.md
docs/product/AI_GOVERNANCE_AND_MODEL_ROUTING.md
docs/product/EDUCATION_OS_RELEASE_GATES.md
docs/product/EDUCATION_OS_RISK_REGISTER.md
docs/product/EDUCATION_OS_BACKLOG.md
```

```
For every claimed domain, classify:

- Native and verified
- Native but incomplete
- Integrated
- Designed/documented only
- Planned
- Not started
- Unsafe to activate
- Requires institutional approval
- Requires legal/security/accessibility review
- Requires migration/reconciliation/rollback design

Do not start implementation until the audit establishes current truth.

============================================================
PHASE 1 — EDUCATION GRAPH FOUNDATION
============================================================

Define and implement or reconcile:

1. Identity graph
2. Institution/campus/school/program/course/section hierarchy
3. Membership and role grants
4. Student-owned workspace graph
5. Academic path graph
6. Course and learning graph
7. Evidence/provenance graph
8. Support and intervention graph
9. Skills/credential/career graph
10. Consent and sharing graph
11. Commercial/customer graph
12. Governance/audit/risk graph

For every graph edge define:

- Source authority
- Tenant scope
- Permission rule
- Data classification
- Retention rule
- Consent requirement
- Audit event
- Integration source
- Migration source
- Native replacement status

============================================================
PHASE 2 — ROLE-SPECIFIC EDUCATION OS
============================================================

Build coherent role experiences:

A. Student OS
- Home
```

- Today
- Action Center
- Path
- Plan
- Calendar
- Tasks
- Notes
- Files
- Search
- Courses
- Study
- Community
- Support
- Career
- Profile
- AI copilot

B. Faculty / Course Studio
- Course home
- Syllabus
- objectives
- resources
- assignments
- study packs
- course guidance
- AI rules
- discussions
- gradebook
- regrade
- feedback
- office hours
- accessibility and integrity tools

C. Advisor / Student Success
- Caseload
- student path
- action completion
- check-ins
- referrals
- support
- success plans
- outcome insight
- privacy-scoped sharing

D. Registrar / Academic Operations
- Terms
- catalog
- sections
- registration
- holds
- overrides
- enrollment
- academic record
- grades
- regrades
- degree progression

```
- graduation
- transfer evaluation

E. Campus Services
- dining
- housing
- events
- organizations
- community
- study spaces
- campus employment
- support offices
- accessibility services
- safety handoffs

F. Family/Guardian
- consented invitations
- scoped sharing
- revocation
- audit history
- resource/support handoff

G. Career / Alumni / Employers
- skills graph
- verified evidence
- portfolio
- opportunities
- mentors
- career advising
- alumni engagement
- lifelong credentials

H. Institutional Control Plane
- policy
- configuration
- workflow builder
- integrations
- AI provider policy
- security
- privacy
- retention
- audit
- approvals
- rollouts
- trust
- compliance
- support
- offboarding

I. Semester Company Operations
- GTM
- pilots
- billing
- customer success
- renewals
- trust
```

```
- compliance
- support
- security
- release operations
- financial visibility
- workforce/capacity planning

============================================================
PHASE 3 — AI EDUCATION PLATFORM
============================================================

Build a governed model-routing architecture.

Support approved external AI models but make Semester the governance and
education context layer.

For each AI interaction require:

- identity
- tenant
- role
- purpose
- course/context scope
- data classification
- consent
- AI provider policy
- course AI rules
- academic-integrity rule
- source/knowledge retrieval policy
- budget/rate limit
- logging/audit
- response provenance
- human escalation option

Build:

- student copilot
- course-aware tutor
- research assistant
- writing assistant
- study planner
- registration guide
- advising support assistant
- staff workflow assistant
- institutional knowledge assistant
- faculty authoring assistant
- AI evaluation system
- AI red-team and safety test plan
- model/provider registry
- spend meter
- policy engine
- source/citation layer
- human override and escalation
- tenant AI configuration
- course AI configuration
```

```
No AI output becomes official institutional truth without explicit authority,
human approval, and audited workflow.

================================================================
PHASE 4 — INTEROPERABILITY AND MIGRATION
================================================================

Implement or define:

A. LTI
- OIDC login
- state/nonce validation
- issuer/audience/deployment validation
- role and course context
- deep linking
- membership binding
- tool launch audit
- controlled grade passback
- consent and course policy integration

B. OneRoster
- CSV staging/import
- REST sync
- users
- orgs
- terms
- courses
- classes
- enrollments
- resources
- line items
- results/grades
- mapping versioning
- soft deletes
- reconciliation
- error queue
- source freshness
- export and rollback

C. Edu-API and enterprise education APIs
- academic enterprise mapping
- standard vs institution-specific adapter model
- source authority
- change detection
- idempotency
- audit
- reconciliation

D. SIS/ERP migration
- inventory
- field mapping
- data classification
- staging
- data quality
- duplicate handling
- reconciliation
```

```
- dual run
- cutover
- rollback
- archive/export
- retention
- offboarding

E. Productivity migration
- Google/Microsoft/Apple import/export
- calendar
- files
- tasks
- notes
- identity
- permissions
- data portability

F. AI provider transition
- external provider registry
- model routing
- fallback
- provider policy
- spend policy
- source restrictions
- data restrictions
- evaluation
- incident handling

============================================================
PHASE 5 — DOMAIN REPLACEMENT GATES
============================================================

For every domain, define replacement gates:

1. Functional parity
2. Data model completeness
3. Accessibility
4. Security and privacy controls
5. RLS/tenant isolation
6. Auditability
7. Migration integrity
8. Reconciliation
9. Dual-run reliability
10. User acceptance
11. Institutional policy approval
12. Support readiness
13. Incident/rollback readiness
14. Export/portability
15. Commercial and implementation readiness

No domain may be marked “native authoritative” without passing all applicable
gates.

============================================================
PHASE 6 — OUTCOME AND MOAT
============================================================
```

```
Build outcome frameworks for:

Students:
- action clarity
- meaningful action completion
- learning progress
- time/effort efficiency
- support access
- confidence
- career readiness

Faculty:
- course clarity
- workload reduction
- feedback cycle
- policy enforcement
- learning insight

Institutions:
- student activation
- retention proxies
- support load
- workflow completion
- data quality
- vendor consolidation
- implementation consistency
- operating cost
- trust/compliance evidence

Semester:
- platform adoption
- renewal
- expansion
- gross retention
- net revenue retention
- implementation capacity
- reliability
- security posture
- AI cost/margin
- migration throughput
- partner ecosystem growth

Return:
- complete product architecture
- build order
- role matrix
- data authority matrix
- integration/migration roadmap
- domain replacement gate matrix
- 12-month implementation plan
- launch readiness plan
- risk register
- engineering backlog
- commercial model
- required team roles
```

```
- test strategy
- operating cadence

Do not stop at a high-level vision. Create implementation-ready plans,
schemas, interfaces, test gates, and traceability artifacts.
```

Recommended execution branches

```
# Establish truth
git checkout -b audit/unified-education-os-baseline

# Foundation
git checkout -b feat/education-graph-foundation

# Student OS and daily workspace
git checkout -b feat/student-os-daily-workspace

# Course Studio and learning
git checkout -b feat/course-studio-learning-os

# AI gateway and copilot
git checkout -b feat/governed-ai-copilot

# Advising/student success
git checkout -b feat/student-success-os

# Registrar/academic operations
git checkout -b feat/registrar-academic-os

# LTI, OneRoster, and migration fabric
git checkout -b feat/education-interoperability-fabric

# Campus life/family/career
git checkout -b feat/campus-life-family-career

# Institutional control plane
git checkout -b feat/institutional-control-plane

# Marketplace/platform ecosystem
git checkout -b feat/education-platform-ecosystem
```

Final strategic framing

The thesis is not:

> “Semester is a better portal.”

The thesis is:

> **Semester is the education operating system that replaces the fragmented infrastructure of learning, academic life, student support, campus services, institutional operations, productivity, and AI with one connected, governed, human-centered platform.**

The practical discipline is:

> **Build everything natively. Integrate responsibly. Replace only with proof. Operate every domain as one system.**

⁂

1. https://www.1edtech.org/standards/lti
2. https://www.1edtech.org/standards/oneroster
3. https://www.1edtech.org/standards/edu-api
4. https://www.nist.gov/itl/ai-risk-management-framework
5. https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
6. https://www.perplexity.ai/search/fb4dd733-3429-4e79-a23f-22ed168705d3
7. https://www.perplexity.ai/search/21fb2488-6e02-4456-8782-d255401fae2b
8. https://www.perplexity.ai/search/f8eed7b1-f091-4697-9199-fa3abfefbb9f

expand further on brand identy algorthims workflowws data flows and ui

Semester needs a unified **brand identity system, algorithmic intelligence system, workflow language, data-flow architecture, and UI grammar**. These must work together so the product feels like one calm, trusted Education OS rather than a set of features stitched together.

The design principle is:

> **Semester should make an institutionally complex world feel personally clear.**

A student should immediately understand: where they are, what matters, what is official, what action they can take, and who can help. The existing Semester direction emphasizes a calm, premium, accessible, low-stimulus interface with progressive disclosure—immediate decision first, explanation second, advanced detail on demand.

Brand identity

Brand position

```
Semester is the education operating system.

It turns fragmented academic information, campus systems, learning tools,
support workflows, productivity tools, and AI into one clear, governed,
human-centered environment.
```

Emotional outcome

Audience	Desired feeling
Student	“I know what to do next.”
Faculty	“This helps me teach clearly without adding administrative burden.”
Advisor	“I can support students without chasing information across systems.”
Registrar	“This respects academic authority, auditability, and correctness.”
CIO/security	“This is controlled, governable, interoperable, and operationally mature.”
Parent/guardian	“I can support without invading the student’s privacy.”
Employer	“This is credible, verified educational evidence—not a scraped profile.”
Semester operator	“I can see, govern, support, and improve the platform safely.”

Brand personality

```
Calm, not sterile
Intelligent, not patronizing
Ambitious, not inflated
Academic, not bureaucratic
Modern, not trendy for its own sake
Warm, not childish
Trustworthy, not vague
Human-centered, not surveillance-centered
Powerful, not overwhelming
```

Brand vocabulary

Use

```
Path
Action
Progress
Evidence
Source
Guide
Workspace
Course Studio
Support
Plan
Clarity
Next step
Governed
Trusted
Connected
Official
Student-controlled
Explainable
Accessible
Institution-ready
```

Avoid

```
Disrupt everything
Revolutionary AI magic
Replace humans
Guaranteed outcomes
Fully automated advising
Perfect personalization
Total surveillance
One-click compliance
All-in-one without explanation
Enterprise-grade without evidence
```

Core brand statements

```
Short:
Semester is the education operating system.

Student:
Semester helps you understand where you are, what matters next,
and how to move forward.

Institution:
Semester connects student action, learning, planning, support,
and institutional operations in one governed platform.

Technical:
Semester is a multi-tenant education platform with role-aware workflows,
policy-governed AI, source authority, evidence, interoperability,
and operational controls.

Trust:
Semester is built to make educational information clearer, more actionable,
more accessible, and more accountable.
```

Visual identity

Visual design doctrine

```
Primary visual feeling:
A calm, high-trust academic workspace.

Not:
A noisy student social app.
A legacy university portal.
A generic corporate dashboard.
A childish EdTech product.
A glowing “AI” landing page.
```

Color system

```
Primary:
Indigo / blue-violet
Meaning:
Clarity, action, trusted intelligence

Neutral:
Slate / near-black / soft white
Meaning:
Calm, focus, readable density

Success:
Emerald
Meaning:
Completed, healthy, confirmed
```

```
Warning:
Amber
Meaning:
Needs attention, review, limited confidence

Critical:
Rose/red
Meaning:
Error, security concern, failed action, urgent issue

Information:
Blue
Meaning:
Source, guidance, context, available action
```

Semantic rules

```
Indigo = user action, focus, selected state, trusted navigation
Green = confirmed completion or healthy state
Amber = caution, review, data freshness issue, pending work
Rose = blocked, failed, unsafe, destructive, high urgency
Gray = context, inactive, historical, unavailable
```

Never use status colors as decoration. Never communicate a critical state with color alone.

Typography

```
Display:
Large narrative statements, landing page hero, major decision context

Heading:
Screen titles, page sections, workflow steps

Subheading:
Cards, modules, grouped content

Body:
Primary explanation and reading content

Body small:
Metadata, source/freshness, secondary context

Label:
Form labels, controls, status labels, section metadata

Mono:
Record IDs, audit IDs, correlation IDs, technical configuration
```

Typography rules

```
- One H1 per page.
- Use a strict heading hierarchy.
- Keep student-facing sentences short.
- Use plain language before specialist language.
- Expand acronyms on first use.
- Avoid large blocks of all caps.
- Use numeric/tabular typography for finance, analytics, and schedules.
- Make source, status, and data freshness readable but secondary.
```

Shape, depth, and motion

```
Corner radius:
Moderate, consistent, not overly rounded or toy-like

Elevation:
Minimal; use border and contrast before shadow

Motion:
Short, purposeful, optional
- page transition: subtle
- modal: direct
- save state: visible
- completion: calm
- loading: non-distracting
- reduced motion: always supported

Illustration:
Use only when it clarifies workflow or reduces anxiety
Do not use decorative filler art to make empty screens seem busy
```

Semester UI grammar

The UI must behave the same across every screen.

Universal page shell

```
┌──────────────────────────────────────────────────────────────┐
| Semester logo | Context | Search | Notifications | Profile   |
├──────────────────────────────────────────────────────────────┤
| Primary navigation                                           |
| Today · Path · Plan · Courses · Workspace · Support · Career |
├──────────────────────────────────────────────────────────────┤
| Page title                                                   |
| What this page helps you do                                  |
| Source/freshness/context                                     |
├──────────────────────────────────────────────────────────────┤
| Layer 1: Immediate decision / most important action          |
├──────────────────────────────────────────────────────────────┤
| Layer 2: Explanation, choices, and supporting context        |
├──────────────────────────────────────────────────────────────┤
```

```
| Layer 3: Advanced detail, settings, history, export, audit   |
└──────────────────────────────────────────────────────────────┘
```

Three-layer disclosure model

Layer	Purpose	Example
Layer 1	Immediate decision	“Your registration opens Tuesday at 8:00 AM. You have one hold.”
Layer 2	Explanation and action	“The hold is from Student Accounts. Here is why it matters and who can help.”
Layer 3	Advanced detail	Source system, policy text, history, export, integration state, audit trail

A student should not need Layer 3 to complete a normal daily action. Operators and staff may need Layer 3 frequently, but still should not be overwhelmed by it initially.

Screen question model

Every Semester screen should answer:

```
Where am I?
What is this screen for?
What matters now?
What can I do?
What is official?
What is stale or uncertain?
What happens if I take this action?
Who can help?
Where can I find deeper detail?
```

Algorithms

Semester’s algorithms should be **explainable, policy-bounded, source-aware, and purpose-limited.** No hidden “black box” should make high-impact educational decisions.

Algorithm inventory

```
1. Next-action ranking
2. Student workload estimation
3. Registration readiness
4. Degree path scenario engine
5. Course/section recommendation
6. Study-plan recommendation
7. Learning-resource recommendation
8. Search ranking
9. Support routing
10. Advisor caseload prioritization
11. Student-success aggregate risk signals
12. Career opportunity matching
13. Mentor matching
14. Community safety ranking
15. Event recommendation
```

```
16. Campus service recommendation
17. Notification priority
18. AI model routing
19. AI source retrieval
20. AI answer confidence/escalation
21. Content moderation assistance
22. Integration health scoring
23. Tenant health scoring
24. Customer health scoring
25. Pilot conversion scoring
26. Revenue forecast
27. AI cost forecasting
28. Anomaly detection
29. Data-quality scoring
30. Migration reconciliation
```

Algorithm governance requirements

Every algorithm must have:

```
Purpose
Owner
Users affected
Decision type
Input data
Excluded data
Source authority
Data classification
Tenant policy
Consent requirement
Output
Confidence/uncertainty
Explanation
Human review path
Appeal/correction path
Bias/fairness review
Monitoring
Evaluation dataset
Version
Audit record
Rollback/deprecation plan
```

1. Next-action ranking

Purpose

Show the student the most useful next action without overwhelming them.

Inputs

```
Official deadlines
Course assignments
Registration windows
Holds
Academic plan
Task commitments
Student-selected goals
Support requests
Course schedule
Institution notices
User preference
```

Never use without explicit policy/consent

```
Private message content
Sensitive health data
Protected characteristics
Financial aid details beyond action need
Accommodation details
Disciplinary records
External browsing history
Unapproved behavioral surveillance
```

Ranking formula

$$\text{Action Priority} = w_d D + w_i I + w_b B + w_g G + w_c C - w_o O$$

Where:

$$D = \text{deadline urgency}$$

$$I = \text{institutional importance}$$

$$B = \text{blocker severity}$$

$$G = \text{student-selected goal relevance}$$

$$C = \text{course/program context relevance}$$

$$O = \text{overload penalty}$$

The overload penalty is essential. Semester should not maximize notifications; it should prioritize meaningful action.

Explainability UI

```
Recommended: Review your registration hold

Why this appears:
- Your registration window opens in 3 days.
- This hold may prevent enrollment.
- Student Accounts is the responsible office.

Source:
Official institution record, updated 2 hours ago.

Next step:
View hold details or contact Student Accounts.
```

2. Workload estimation

Inputs

```
Credits
Course meeting schedule
Assignment due dates
Student-entered time estimates
Course workload guidance
Assessment dates
Known schedule conflicts
Student preference
```

Output

```
Low / balanced / heavy workload estimate
Peak weeks
Conflict warnings
Suggested study blocks
Suggested sequencing
```

Important rule

```
Workload estimate is guidance, not academic authority.
Students can override personal estimates.
No punitive institutional action uses workload score.
```

3. Registration readiness algorithm

```
Readiness =
registration window known
+ account status known
+ holds checked
+ plan exists
+ prerequisite status checked
+ credit load reviewed
+ course sections selected
+ alternatives considered
+ official handoff available
```

Output:

```
Ready
Almost ready
Blocked
Needs advisor review
Information unavailable
```

Every readiness item must show its source, timestamp, and action path.

4. Degree path scenario engine

```
Student selects:
- program goal
- term preference
- possible courses
- workload preference
- cost preference
- graduation target

Semester computes:
- requirement coverage
- prerequisite sequence
- credit load
- estimated graduation timeline
- potential conflicts
- missing official information
- advisor/registrar confirmation needs
```

The system must label:

```
Official requirement
Planning estimate
Assumption
Unavailable data
Advisor/registrar confirmation required
```

5. Search ranking

```
Query
  ↓
Permission filter
  ↓
Tenant/course/context filter
  ↓
Source authority ranking
  ↓
Freshness ranking
  ↓
User context relevance
  ↓
Result type diversity
  ↓
Explainable result list
```

Search result categories

```
Official institution policy
Course content
Student-owned content
Approved learning source
Campus service
Support resource
Career opportunity
Community content
External web/research
AI-generated answer
```

Search must never make community content appear more authoritative than official policy.

6. AI model router

```
Request
  ↓
Data classification
  ↓
Tenant provider policy
  ↓
Course AI policy
  ↓
Cost/latency requirement
  ↓
Required capability:
- reasoning
- research
- writing
- code
- retrieval
- summarization
```

```
  ↓
Approved model/provider
  ↓
Fallback model or human handoff
```

Router output

```
Chosen provider/model
Why it was chosen
What context was shared
What sources were used
Whether data was retained
Cost/usage impact
Fallback behavior
```

7. Support routing

```
Student request
  ↓
Issue category
  ↓
Urgency
  ↓
Institution/tenant
  ↓
Student-selected context
  ↓
Office/service ownership
  ↓
Eligibility/policy check
  ↓
Correct queue
  ↓
Escalation path
```

No algorithm should make a clinical, disciplinary, emergency, or financial decision. It can route to an approved official process.

8. Tenant health algorithm

```
Tenant Health =
commercial
+ implementation
+ activation
+ workflow completion
+ support
+ reliability
+ integration health
+ security/compliance
```

```
+ executive engagement
+ renewal status
```

Output must explain:

```
Health band
Score components
Top drivers
Data freshness
Recommended actions
Owner
Due date
```

Workflow architecture

Workflow definition model

Every workflow needs:

```
type SemesterWorkflow = {
  id: string;
  version: number;
  tenantId: string | null;

  name: string;
  purpose: string;
  domain: string;

  trigger: string;
  actors: string[];
  requiredCapabilities: string[];

  inputs: string[];
  outputs: string[];

  sourceAuthority: string[];
  dataClassification: string[];

  states: string[];
  transitions: string[];

  approvalRequirements: string[];
  auditEvents: string[];

  sla?: string;
  supportOwner: string;

  rollbackPlan?: string;
  retentionPolicy?: string;

  accessibilityRequirements: string[];
  aiPolicyRequirements?: string[];
```

```
  successMetrics: string[];
};
```

Universal workflow state pattern

```
Draft
→ Submitted
→ Validating
→ Waiting for information
→ Waiting for approval
→ Approved
→ Executing
→ Completed
→ Failed
→ Escalated
→ Cancelled
→ Archived
```

Workflow examples

```
Student registration
Support request
Advisor referral
Pilot implementation
Integration activation
Data migration
Feature rollout
Price exception
Security document request
Privacy erasure request
Family grant
AI agent action
Course publication
Grade change
Payment plan
Community moderation
Incident response
Tenant offboarding
```

Data-flow architecture

Data flow standard

```
Source system
  ↓
Connection approval
  ↓
Scope approval
  ↓
Secure ingestion
  ↓
```

```
Staging
  ↓
Validation
  ↓
Transformation
  ↓
Canonical model
  ↓
Source/provenance record
  ↓
Policy and permission filter
  ↓
Role-specific read model
  ↓
User interface
  ↓
Action/command
  ↓
Audit/event/outbox
  ↓
Projection/analytics
```

Data layers

```
Raw external payload
  ↓
Staging data
  ↓
Validated mapped data
  ↓
Canonical education graph
  ↓
Domain workflow records
  ↓
Read models/projections
  ↓
User interface
  ↓
Analytics/aggregate outcomes
  ↓
Export/archive/deletion
```

Data classification

Tier	Example	Handling
T0	Public marketing content	Public cacheable
T1	Internal operational metadata	Authorized staff
T2	Student-owned workspace data	Student/private by default
T3	Institutional academic/personal data	Tenant-scoped, policy controlled

Tier	Example	Handling
T4	Sensitive support/accessibility/financial data	Restricted, purpose-limited, audit-heavy
T5	Credentials, secrets, authentication material	Server-only, never browser exposed
T6	Highly restricted legal/security evidence	Controlled access, named reviewers only

UI system

Required shared components

```
AppShell
ContextBar
GlobalNavigation
MobileNavigation
GlobalSearch
CommandPalette
PageHeader
SourceBadge
FreshnessBadge
AuthorityBadge
PolicyBadge
ConsentBadge
StatusBadge
HealthBadge
RiskBanner
ActionCard
ActionCenter
WorkflowStepper
DecisionCard
EvidenceDrawer
AuditTimeline
DataTable
EmptyState
LoadingState
ErrorState
OfflineState
ForbiddenState
SupportHandoff
AIAnswerPanel
CitationPanel
ActionPreview
ApprovalBanner
FormField
ErrorSummary
Modal
Toast
NotificationCenter
```

Standard data trust panel

Every critical screen should show:

```
Source: Official institution record
Updated: 2 hours ago
Authority: Registrar
Status: Current
Policy: Institution registration policy v4.2
Action: View official details
```

Standard AI panel

```
Answer
Sources
What this means
Suggested next step
Limits and uncertainty
Course/institution policy
Give feedback
Ask a person / official office
```

Standard error state

```
What happened
What data may be affected
What you can do now
Whether action was saved
Whether it will retry
Who can help
Reference ID
```

Master design/data/algorithm execution prompt

```
Build the Semester Brand, Intelligence, Workflow, Data-Flow, and UI System.

Semester is the unified Education Operating System. It must feel like one calm,
premium, accessible, trustworthy product across student, faculty, advisor,
registrar, campus, family, career, institutional, company, and developer
experiences.

Create:

docs/design/SEMESTER_BRAND_IDENTITY.md
docs/design/SEMESTER_VISUAL_LANGUAGE.md
docs/design/SEMESTER_UI_GRAMMAR.md
docs/design/SEMESTER_COMPONENT_LIBRARY.md
docs/design/SEMESTER_INTERACTION_LANGUAGE.md
docs/design/SEMESTER_MOTION_AND_FEEDBACK.md
docs/design/SEMESTER_DATA_TRUST_UI.md
```

```
docs/design/SEMESTER_RESPONSIVE_STRATEGY.md
docs/algorithms/SEMESTER_ALGORITHM_CATALOG.md
docs/algorithms/SEMESTER_ALGORITHM_GOVERNANCE.md
docs/algorithms/SEMESTER_NEXT_ACTION_RANKING.md
docs/algorithms/SEMESTER_WORKLOAD_MODEL.md
docs/algorithms/SEMESTER_REGISTRATION_READINESS.md
docs/algorithms/SEMESTER_SEARCH_RANKING.md
docs/algorithms/SEMESTER_AI_ROUTING.md
docs/workflows/SEMESTER_WORKFLOW_ARCHITECTURE.md
docs/workflows/SEMESTER_WORKFLOW_CATALOG.md
docs/data/SEMESTER_DATA_FLOW_ARCHITECTURE.md
docs/data/SEMESTER_DATA_CLASSIFICATION.md
docs/data/SEMESTER_SOURCE_AUTHORITY.md
docs/data/SEMESTER_PROVENANCE_MODEL.md

For every algorithm define:
- purpose
- owner
- affected users
- inputs
- excluded inputs
- source authority
- tenant scope
- data classification
- consent
- output
- explanation
- confidence
- human escalation
- bias/fairness review
- evaluation
- monitoring
- version
- audit
- rollback

For every workflow define:
- trigger
- actors
- capabilities
- states
- transitions
- approvals
- audit
- inputs/outputs
- support owner
- SLA
- retention
- rollback
- success metric

For every UI component define:
- semantic token usage
- accessibility requirement
- responsive behavior
- loading/empty/error/offline/forbidden states
```

```
- analytics event
- source/freshness display
- test coverage

The product must use progressive disclosure:
Layer 1: immediate decision
Layer 2: explanation and action
Layer 3: advanced detail and history

Do not use opaque high-impact algorithms.
Do not use protected or sensitive data for recommendations without explicit
authority.
Do not treat AI output as official institutional truth.
Do not use visual decoration instead of clarity.
Do not allow modules to invent their own colors, spacing, status vocabulary,
permission behavior, or source labels.

Return:
- complete brand system
- complete algorithm inventory
- workflow map
- data-flow map
- UI component system
- screen consistency plan
- accessibility plan
- analytics plan
- testing plan
- P0/P1/P2 backlog
- exact next 25 implementation actions.
```

The final design principle

```
One brand language
One source-authority language
One action language
One policy language
One privacy language
One error-recovery language
One accessibility standard
One workflow grammar
One education graph
= One Semester OS
```

⁂

1. projects.semester.cross_device_experience
2. https://www.perplexity.ai/search/c9a3bc23-faef-4a39-b797-d8dd40ace80f
3. https://www.perplexity.ai/search/7249ab05-f244-41ee-9573-13192050bbd6
4. projects.semester.branding
5. https://www.perplexity.ai/search/ac312ed1-3dc7-4597-b9d0-63678f97be71
6. https://www.perplexity.ai/search/170323cf-9154-442a-b80c-7fc5fd712889

7. https://www.perplexity.ai/search/e8fa66a2-9133-47dc-b9e6-912f6d4ed810